Measuring the Gap: Semantic Alignment Between AI-for-Sustainability Research and SDG Policy Frameworks

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Contents

Abstract

The Sustainable Development Goals (SDGs) provide the dominant international framework for sustainability governance, yet shared SDG labels do not necessarily imply shared semantic framing between research and policy-facing discourse. Existing bibliometric studies map AI research to SDGs using keyword or classification methods, but rarely compare the semantic content of research and international institutional SDG policy discourse. This study uses Sentence-BERT embeddings and a nearest-centroid SDG instrument validated against expert-labelled SDG benchmarks (macro-F1 = 0.716) to construct a diagnostic map of observed semantic distance between 2,543,698 AI-for-sustainability research title-abstract texts and 54,082 segments drawn from 2,456 international institutional SDG policy sources. The analysis distinguishes coverage gaps, which capture which SDGs each corpus emphasises, from within-SDG semantic gaps, which capture how far research and policy texts assigned to the same SDG sit apart in embedding space. Because the corpora also differ in language style and communicative purpose (what linguists call register), these gaps are interpreted as observed textual-semantic divergence rather than as a cleaned measure of substantive disagreement.

The results show distinct patterns. First, institutional policy discourse is concentrated on SDGs 13, 17, and 16, while research concentrates on SDGs 3, 4, and 9. Second, the largest within-SDG semantic gaps occur in SDGs 17 and 13, with SDGs 6, 10, and 7 also among the highest-gap cases. Third, research attention alone does not predict semantic divergence ($r = -0.161$, $p = 0.536$ (probability of observing this correlation by chance); $n = 17$), even though larger cross-corpus coverage imbalance tends to coincide with larger semantic gaps. These findings do not prove substantive policy misalignment. Rather, they show that shared SDG labels are insufficient evidence of shared framing and that embedding-based distance can be used as a transparent diagnostic map of semantic proximity and divergence. Appendix diagnostics further show that the results should be interpreted with caution because of register effects, policy source-family heterogeneity, OpenAlex retrieval boundaries, and an SDG 4 assignment that should be treated as artefact-affected rather than as a clean education finding.

1 Introduction

Artificial intelligence is now routinely presented as part of the solution to sustainable development. In academic research, papers increasingly claim that a model, application, or dataset contributes to one or more Sustainable Development Goals (SDGs), with SDG attribution often functioning as a form of relevance-signalling beyond technical novelty. In international institutional SDG policy discourse, international organisations, intergovernmental bodies, and national AI strategies likewise frame AI as an enabler or accelerator of SDG progress (OECD, 2019; UK Government, 2021; UN AI Advisory Body, 2024; UNESCO, 2021; United Nations, 2015). Across these domains, a shared narrative has taken hold: AI is expected to help deliver the global development agenda.

That narrative matters because it shapes research framing, funding priorities, and policy expectations. Yet the apparent alignment it implies is usually inferred rather than demonstrated. Researchers and policymakers may converge on the same SDG labels while emphasising different problems, actors, timescales, and interventions. A paper and a policy document can both invoke SDG 3 (Good Health) or SDG 13 (Climate Action) without addressing the same substantive terrain. High-level thematic agreement can therefore mask deeper divergence. If so, current claims about "AI for SDGs" may overstate how far research effort is aligned with policy priorities, not because the narrative is necessarily false, but because the methods used to validate it remain too coarse.

The underlying problem is not new. Questions of research-policy divergence have been a central concern in policy studies and knowledge-utilisation scholarship for decades. What is new is the combination of this longstanding problem with the rise of AI-for-SDG framing and the availability of embedding-based semantic tools that make large-scale measurement possible. These tools allow broad discourse-level proximity to be assessed as an empirical property of text, rather than inferred only from anecdote, keyword coincidence, citation or mention counts, or aggregate coverage measures. In that sense, the aim of this paper is deliberately specific: to map how closely academic AI-for-sustainability research and international institutional SDG policy discourse occupy nearby regions in semantic space.

A substantial body of bibliometric work documents *what* AI research engages with: studies consistently find concentration on technically tractable SDGs such as SDG 3, SDG 7, SDG 9, and SDG 13, with relative neglect of SDGs 5, 10, 16, and 17 (Cowls et al., 2021; Singh et al., 2024; Vinuesa et al., 2020). However, most of this literature maps research to the SDGs through keywords, topic counts, expert judgements, or bibliometric assignment. These approaches are valuable for measuring coverage, but they implicitly assume that shared SDG labels imply alignment. They can indicate which goals are mentioned or assigned; they cannot tell whether research and policy mean similar things when they invoke the same goal. Nor do they test whether distributional overlap and semantic overlap move together or come apart. Shared labels, in short, do not by themselves establish shared meaning.

This study addresses that gap. Its central claim is straightforward: shared SDG labels are insufficient evidence of shared framing. I make three contributions. First, I advance AI–SDG mapping from Level 1 (lexical or bibliometric correspondence) to Level 2 (topical proximity in semantic space), using Sentence-BERT (Reimers & Gurevych, 2019) embeddings and SDG centroids derived from the OSDG Community Dataset (Pukelis et al., 2022). Second, I distinguish between a *coverage gap* (which SDGs each corpus emphasises) and a *semantic gap* (how differently the two corpora discuss the same SDG), and measure both across all 17 SDGs. Third, I evaluate whether these two dimensions are related, or whether apparent overlap in SDG coverage can coexist with substantial semantic divergence.

The central research question is therefore: *To what extent do academic AI-for-sustainability research and international institutional SDG policy discourse differ in SDG coverage and semantic framing across the 17 SDGs, and how are these two dimensions related?*

In brief, the two corpora differ substantially in both what they prioritise and how they frame those priorities, and the pattern of divergence is informative.

The remainder of this paper is structured as follows. Section 2 reviews the relevant literature on AI-SDG mapping, research-policy alignment, and semantic similarity methods. Section 3 describes the corpora, measurement instrument, and gap analysis procedures. Section 4 presents the empirical findings. Section 5 interprets the findings through the productive/problematic misalignment distinction and discusses limitations. Section 6 concludes with implications and directions for future research.

2 Literature Review

2.1 AI and the SDGs: What Bibliometrics Shows

Most existing studies of AI and the SDGs ask which goals attract the most research attention, but they have not compared what research and policy texts actually say within each goal. Vinuesa et al. (2020) conducted the most widely cited study, in which an expert panel assessed how AI could enable or inhibit each of the 169 SDG targets; their finding that AI could facilitate 134 targets while potentially inhibiting 59 has shaped discourse around AI governance. However, as the authors acknowledge, the study reflects expert perception rather than empirical measurement of the research literature.

Broader SDG scientometric work has already mapped the growth, institutional distribution, thematic structure, and interlinkages of MDG/SDG-related research. For example, Bautista-Puig et al. (2021) identify global MDG/SDG-related research from 2000–2017 using core publications and citation expansion, then classify outputs into SDGs using a glossary-based approach. Subsequent AI-specific empirical work has mapped the AI publication record to the SDGs. Singh et al. (2024) analysed AI-for-SDG publications over a 20-year period and found that SDGs 3 and 7 were the areas with the most AI applications. Cowls et al. (2021) examined 108 deployed AI for Social Good projects and found SDG 3 dominant while SDGs 5, 16, and 17 were largely unaddressed. Nedungadi et al. (2024) applied topic modelling to a broader corpus and confirmed these patterns. The methodological concern raised by Armitage et al. (2020) is important: SDG publication mapping depends on interpretive choices about what counts as SDG relevance, how target concepts are translated into queries or labels, and which bibliographic database is used. The embedding-based approach used here reduces some vocabulary-level brittleness of keyword dictionaries, but it remains a distributional measurement strategy whose results are conditional on corpus construction, centroid provenance, and model choice. Later benchmarking work reaches the same caution from a broader comparison of Boolean, machine-learning, and hybrid SDG-mapping systems: Kashnitsky et al. (2024) find that no single approach performs best across all validation datasets, and that no single "golden" SDG validation set exists.

Not only are SDG mapping systems methodologically contingent; the SDGs themselves are interdependent by design. Nilsson et al. (2016) observe that "the goals depend on each other" but that no operational framework specifies how, warning that treating targets in isolation "risk[s] perverse outcomes." Single-SDG assignment is therefore a simplifying assumption, not a reflection of how goals operate in practice.

One SDG is structurally distinct from this interdependence pattern. Le Blanc (2015), in a network analysis of the proposed SDGs, characterises SDG 17 (Partnerships for the Goals) as the "dedicated goal" for "cross-cutting means of implementation for the whole set of SDGs" — encompassing finance, technology transfer, capacity-building, and multi-stakeholder partnerships — and excludes it from the cross-goal network mapping because its targets are not substantive domains but implementation infrastructure, a distinction without parallel among the other 16 goals. The Griggs et al. (2017) guide to SDG interactions similarly describes SDG 17 as providing "the main enablers for implementing the entire SDG framework" across five categories: finance, technology, capacity-building, trade, and systemic issues including policy coherence and partnerships. Unlike every other SDG, where both research and policy discourse centre on a shared substantive topic, SDG 17 policy discourse concerns the coordination and resourcing of the implementation apparatus itself — a conceptual space that research abstracts on technical AI applications are not designed to inhabit.

Despite convergent findings on which SDGs attract the most research, these bibliometric and scientometric studies share two limitations relevant to this paper. Even when they move beyond simple keyword searches through citation expansion, topic modelling, or glossary-based SDG assignment, or hybrid query–machine-learning systems, they primarily measure research coverage and interlinkages rather than within-SDG semantic framing. They also make little direct comparison to the policy-facing documents through which the SDG framework is translated into action. Where policy documents have been used in adjacent scientometric work, they have more often served as sources of mentions or citations rather than as full-text corpora for comparing semantic framing (Bornmann et al., 2016).

2.2 Research-Policy Alignment in AI and Sustainability

A growing literature documents structural tensions between what AI research produces and what sustainability policy requires. Strauss et al. (2025) show that corporate AI research concentrates on pre-deployment technical problems, creating a gap between research priorities and the real-world deployment impacts that governance and policy must address. Sioumalas-Christodoulou and Tympas (2025) find that global AI metrics focus on technical and economic performance while policy discourse emphasises ethical and social dimensions aligned with SDG-related concerns. Toney-Wails et al. (2024) provide a close methodological precedent in the FAccT context, showing that policy and research communities can use shared trustworthy-AI terminology while differing in term salience, definitions, and focal concerns.

McCarthy et al. (2026) report a parallel finding in conservation science, where a multi-target SciBERT classifier applied to Ross Sea MPA research reveals that 59% of policy-prioritised research topics appear in fewer than 15 papers, demonstrating that research-policy coverage gaps are observable across domains.

Adjacent scientometric work has also used policy documents to measure research-policy connection, but at the level of mentions rather than semantic framing. Bornmann et al. (Bornmann et al., 2016) examine climate-change publications in Altmetric-tracked policy documents and find that only 1.2% of 191,276 publications had at least one policy mention. Their study is useful here as a boundary marker: policy-document mentions can indicate one narrow form of policy uptake, but they do not show how research is discussed, interpreted, or framed inside policy discourse. The present study therefore addresses a different layer of the research-policy relationship: not whether policy documents cite particular papers, but whether research and policy-facing texts assigned to the same SDG occupy nearby regions in semantic space.

A related research-evaluation literature treats the research-policy relationship through the lens of societal impact measurement. Bornmann (Bornmann, 2017) argues that measuring societal impact is harder than measuring scientific impact because the target of impact must first be specified: research may affect policymakers, business, culture, public health, or other social domains. Policy documents are therefore useful as one possible target-oriented source for measuring impact on policymaking, but this is a different task from semantic framing comparison. The present study does not ask whether research has achieved policy impact; it asks whether research and policy-facing texts occupy similar semantic regions when organised through shared SDG labels.

These findings are consistent with theoretical accounts of research-policy misalignment. Heink et al. (2015) identify credibility, relevance, and legitimacy as determinants of whether scientific knowledge informs policy; topical relevance alone is insufficient — relevance also depends on whether research results can shape actual decisions. In sustainability science, Cash et al. (2003) make a closely related argument: knowledge systems are more likely to support action when they produce information that is simultaneously salient, credible, and legitimate, and when they manage the boundary between knowledge and action through communication, translation, and mediation. The epistemic communities framework (Haas, 1992) provides a theoretical reason why shared framing matters for policy coordination under uncertainty: expert communities can influence policy by articulating cause–effect relationships, framing issues for collective debate, proposing policies, and identifying salient points for negotiation. However, Haas's concept is broader than semantic proximity alone, involving shared normative beliefs, causal beliefs, validity criteria, and a common policy enterprise. The present study therefore does not operationalise an epistemic community directly; it measures the weaker condition of observed semantic proximity between research and policy-facing discourse.

Cross (2013) strengthens this caution by arguing that epistemic communities should not be treated as simply present or absent: their influence varies with internal cohesion, professionalism, access to decision-makers, competition with other actors, and broader political context. This means that semantic proximity can at most indicate a possible precondition for knowledge transfer, not the existence, strength, or influence of an epistemic community.

The SDG framework is itself a politically negotiated instrument with documented limitations. Fukuda-Parr (2016) argues that global goals can have "distorting effects, redefining the meaning of development, and shaping policy by creating incentives to over-focus on target achievement." Bexell and Jönsson (2017) find that SDG summit documents remain "state-centric with great room for state sovereignty," avoid assigning causal responsibility for structural problems, and rely on "weak and unsystematic accountability."

A more fundamental tension exists within the framework itself. Hickel (2019) demonstrates that "Goal 8 violates the sustainability objectives of the SDGs" — 3% annual global growth is empirically incompatible with the resource-use and emissions targets in Goals 6, 12, 13, 14, and 15. The SDGs "lack theoretical grounding" and embody a contradiction between growth and ecological sustainability.

Measuring semantic alignment with SDG centroids therefore uses a benchmark that is itself politically negotiated and internally inconsistent — not a neutral standard. The Gibbons observation that Mode 1 research follows disciplinary rather than policy agendas therefore operates alongside a further ambiguity: what it means to be "aligned" with a framework whose component goals are in tension with each other.

The two-communities theory (Caplan, 1979) provides a complementary explanation grounded in empirical observation: researchers and policy makers "live in separate worlds with different and often conflicting values, different reward systems, and different languages," making divergence the default condition of research-policy interaction rather than a failure needing remediation.

Gibbons et al. (1994) distinguish Mode 1 knowledge production, where problems are set and solved within academic disciplinary interests, from Mode 2, where problems arise in contexts of application; to the extent that research follows Mode 1, its agendas may reflect internal disciplinary dynamics rather than external policy needs. This literature also suggests that divergence is not uniformly problematic: where research explores possibilities that policy discourse has not yet absorbed, distance may reflect productive exploration rather than failure of alignment — a distinction the present study returns to explicitly below.

2.3 Semantic Similarity Methods for Cross-Corpus Comparison

The use of Sentence-BERT (Reimers & Gurevych, 2019) and related transformer-based encoders for cross-corpus semantic comparison has been validated across several domains relevant to this study. This method follows the distributional hypothesis that meaning can be described through patterns of linguistic co-occurrence, a principle whose modern computational implementations trace back to Harris (1954)'s foundational observation that the distribution of a linguistic element is "the sum of all its environments." Sentence-BERT and related transformer-based encoders are therefore used here as pragmatic tools for large-scale semantic comparison, while the interpretation of distance remains descriptive rather than causal. Gjorgjevikj et al. (2025) benchmarked sentence encoders specifically for SDG association tasks, finding that general-purpose models from the sentence-transformers family perform competitively as baselines, and that domain-specific fine-tuning further improves predictive performance while reducing sensitivity to description length.

Rodriguez et al. (2023)'s conText framework provides a closer computational-social-science precedent for comparing meaning across contexts. Their embedding-regression approach estimates how focal terms are used differently across covariates such as party, time, or corpus, while explicitly treating embedding-based "meaning" as a thin distributional description rather than a causal model of human interpretation. This study does not adopt their word-level regression estimator, but it follows the same broad premise that semantic representations can be used descriptively to compare how different discourse communities frame shared terms or categories.

An older corpus-linguistic tradition gives a further reason for caution when interpreting cross-corpus semantic distance. Biber's multidimensional analysis of spoken and written English showed that register differences are not simple binary contrasts, but continuous patterns of co-occurring linguistic features across genres (Biber, 1988). For the present study, this matters because research abstracts and policy documents may differ not only in substantive topic, but also in abstraction, modality, institutional stance, and rhetorical function. Embedding distance is therefore treated as a measure of observed textual-semantic proximity, not as a purified measure of topic alone.

The OSDG Community Dataset (Pukelis et al., 2022) provides one of the labelled corpora from which the SDG centroids are derived. The dataset contains 37,575 texts validated for SDG relevance by a global community of annotators, covering SDGs 1–16 (SDG 17 is absent from the dataset). This absence is not a result of data processing choices made here; it is a documented property of the OSDG Community Dataset itself: the OSDG team report that efforts for SDG 17 were "halted due to the complexity involved in analysing it" (see Pukelis et al., 2022, §3.5). OSDG labels are not simply machine-generated labels: volunteers are shown a text and a suggested SDG label, then accept or reject that label, with agreement recorded across multiple annotators. This makes OSDG suitable as a broad human-validated reference corpus, while also introducing a limitation: because annotators validate a suggested label rather than freely choosing among all 17 SDGs, the labels partly inherit the framing of the source documents and the original suggested labels. Because many OSDG source texts and suggested labels originate from UN-adjacent SDG document collections, the centroids may be calibrated toward policy vocabulary — a potential bias the analysis tests for explicitly. High-quality graded relevance sets for SDG classification do not exist, making community-validated datasets such as OSDG the current best available resource (Kashnitsky et al., 2024).

To supplement OSDG's missing SDG 17 coverage and to broaden the reference base for all goals, three additional labelled corpora are introduced. The IISD (International Institute for Sustainable Development) SDG Knowledge Hub (Wulff et al., 2023) provides 616 single-label SDG 17 texts (out of 2,221 single-label texts across SDGs 1–17 from 9,172 multi-labelled news articles), offering an independent journalistic-annotation perspective. The SDGi Corpus (SDGi, 2024) contributes 321 single-label SDG 17 texts (out of 5,233 single-label texts across SDGs 1–17 from national Voluntary National Review (VNR) / Voluntary Local Review (VLR) reports), providing a within-genre policy-language reference. The Aurora dataset (Vanderfeesten et al., 2020) contributes 5,619 expert-validated single-label texts (SDGs 1–17, 107 for SDG 17), covering all 17 goals with independent expert annotations of scientific publications. The SDG Classification Benchmark (SDG Classification Expert Group, 2025) is used exclusively as an independent validation set and does not contribute to any centroid.

The nearest-centroid approach — computing the mean embedding of labelled texts for each SDG and using cosine distance to the nearest centroid as the classification criterion — is conceptually simple, computationally efficient, and avoids the risk of overfitting that characterises supervised classifiers trained on small labelled sets. This simplicity is a deliberate methodological choice rather than a fallback: the study requires a transparent measurement instrument for corpus-level comparison, not an opaque operational classifier. A fine-tuned classifier might improve individual-document accuracy, but it would also introduce additional training choices, hyperparameters, and overfitting risks that are unnecessary for the present purpose. The centroid represents the "direction" in semantic space most characteristic of each SDG, making the resulting score vectors directly interpretable.

2.4 Theoretical Framing: Productive and Problematic Misalignment

Not all research-policy misalignment is necessarily harmful. Heilinger et al. (2024) warns that the "sustainable AI" label can be used for greenwashing where technologies are framed as contributing to the SDGs without sufficient basis. Heras-Saizarbitoria et al. (2022), analysing 1,370 sustainability reports, find that most organisations' SDG engagement is superficial — "the SDGs only serve to add colour and fancy icons to reports" — suggesting that SDG-washing is a measurable empirical pattern. This supports the caution that reported alignment should not be conflated with substantive commitment. But the converse risk is also real: demanding that research track policy discourse too closely could suppress basic research and frontier exploration that creates future policy options. Building on the broader research-policy literature, and in the spirit of Toney-Wails et al. (2024), I therefore distinguish between *productive misalignment* — where research explores SDG-relevant possibilities that policy discourse has not yet absorbed — and *problematic misalignment* — where research and policy-facing discourse remain distant in areas that are already institutionally salient or policy-dominant. This distinction is interpretive rather than measured: embedding analysis can identify where divergence occurs but cannot classify it as productive or problematic without qualitative follow-up. The conceptual contribution is therefore diagnostic: it shifts the question from whether research and policy are simply "aligned" or "misaligned" to where divergence is visible and what kinds of follow-up would be needed to interpret it.

2.5 Research Gap

To the best of my knowledge, the literature reviewed above has not produced: (i) a systematic within-SDG semantic gap analysis comparing what research and policy-facing discourse say *about the same SDG*, rather than measuring research coverage, citation patterns, or policy-document uptake; (ii) a bidirectional alignment diagnostic — though the present study is unable to separate genre effects from substantive signal for this quantity and reports it only as an appendix sensitivity check rather than as a headline result; or (iii) an examination of whether coverage gaps and semantic gaps are positively related, weakly related, or inversely related. These are the contributions of the present study. Together, they provide a systematic framework for distinguishing research-policy coverage from research-policy semantic proximity within a shared SDG representation.

3 Methodology

3.1 Conceptual Framing: Topical Overlap as a Measurable Proxy

This study measures *topical overlap* between research and policy-facing discourse — whether the two corpora discuss the same subject areas, as proxied by semantic similarity in embedding space. I do not claim to measure substantive alignment (whether research and policy propose compatible solutions to the same problems), causal influence (whether research shapes policy or vice versa), or operational alignment (whether the same methods and interventions are implied). Topical overlap is the necessary but not sufficient condition for research-policy dialogue: shared vocabulary does not guarantee shared meaning or direction, but its absence makes dialogue structurally impossible. This study should therefore be read as a diagnostic analysis of observed semantic distance, not as a causal or normative proof of substantive policy misalignment.

Nor should the measure be interpreted as a societal-impact metric. Unlike policy-document mention studies (Bornmann et al., 2016), this design does not track whether specific papers are cited or used by policy actors — it measures observed textual-semantic proximity between two corpora, not shared causal beliefs in the full epistemic-community sense (Haas, 1992).

In this sense, the analysis follows a thin distributional view of meaning: embedding distances describe differences in textual context and usage, not direct effects on human understanding or policy uptake. The distinction is standard in distributional semantics evaluation: the SemEval SICK task (Marelli et al., 2014) evaluates systems separately on semantic relatedness (predicting the degree of similarity between sentences) and textual entailment (determining whether one sentence follows from another), confirming that the two are treated as distinct evaluation dimensions.

I use the term register in the corpus-linguistic sense: a text variety associated with a situation of use and communicative purpose, rather than merely a superficial style label. Biber and Conrad distinguish register from genre and style: register analysis relates pervasive linguistic features to situational context and communicative function, while genre analysis focuses more on conventional whole-text structures and style on aesthetic or authorial preferences (Biber & Conrad, 2009). This distinction matters here because the research-policy contrast is primarily a register contrast: research abstracts and policy documents are produced in different institutional settings, for different audiences, and for different communicative purposes.

This framing positions this contribution at Level 2 of a three-level hierarchy: Level 1 measures lexical or bibliometric correspondence, Level 2 measures topical similarity in embedding space, and Level 3 would require structured comparison of implied problems, actors, interventions, and implementation pathways. Toney-Wails et al. (2024) illustrate why shared terminology may still conceal deeper divergence, but full Level 3 operational alignment remains beyond the scope of this study. Advancing to Level 2 allows me to detect semantic divergence that keyword methods miss, while avoiding claims about implementation pathways that unsupervised text analysis cannot support.

A further conceptual note is necessary on the linguistic register difference between corpora. Following corpus-linguistic work on register variation, I treat the research-policy contrast not as a minor stylistic nuisance, but as a comparison between text varieties that may differ along empirically recoverable linguistic dimensions (Biber, 1988). Policy documents use performative modal language ("must", "should", "will") and articulate goals at high abstraction. Research papers use hedged indicative language ("we demonstrate", "results suggest") and report empirical findings at high specificity. High cosine similarity between a research abstract and a policy segment therefore indicates that they discuss nearby textual-semantic terrain, not that they are "saying the same thing." Conversely, low cosine similarity may reflect substantive topical divergence, register divergence, or both. All findings in this paper are framed accordingly.

(Grieve et al., 2025) formalise this logic at the level of language models, defining a variety of language as a population of texts delimited by extra-linguistic factors—especially dialect, register, and period—and arguing that any model trained on a corpus is inherently modelling the variety of language that corpus represents. Under this view, Sentence-BERT's embedding space is not a neutral semantic geometry. It inherits the variety structure of the corpus on which the encoder was pre-trained, and when two sub-populations of texts (research abstracts and policy segments) are projected into that space, their observed distance reflects the degree to which they participate in distinct varieties of language, not topic alone. The semantic gap reported in this study is therefore a measure of observed corpus-level varietal divergence, not a purified measure of substantive disagreement.

3.2 Research Corpus

The research corpus consists of 2,543,698 English-language research abstracts retrieved from the OpenAlex API (Priem et al., 2022), covering publications between 2018 and 2025. Abstracts were retrieved using a structured query combining OpenAlex's native

SDG work filter with four AI-related search terms ("artificial intelligence", "machine learning", "deep learning", "neural network"), yielding 68 queries: 17 SDGs times four AI terms.[1] This makes OpenAlex the retrieval mechanism rather than the final labelling authority. OpenAlex SDG metadata defines the candidate universe of AI-for-sustainability papers; the SDG assignments used in the analysis are produced later by the independently validated centroid instrument described below. Following Armitage et al. (2020), this retrieval universe should be treated as a method-dependent corpus boundary rather than a neutral population of all SDG-relevant AI research. After deduplication by OpenAlex work ID and removal of records lacking usable abstracts, 2,543,698 unique papers were retained in the canonical corpus. No balancing by SDG, citation count, or OpenAlex relevance rank is applied, because such trimming would distort the coverage profile the study aims to measure. Each paper's title and abstract were concatenated to form the text unit for embedding.

The 2018 start date reflects the post-AlexNet emergence of deep learning as the dominant AI paradigm and the adoption of the SDG framework in 2015 with a research uptake lag. Earlier papers are excluded to ensure comparability in methodological framing.

3.3 Policy Corpus

The policy corpus contains 54,082 text segments drawn from three source collections, representing 2,456 unique source documents. Substantively, it should be read as mixed *international institutional SDG policy discourse*, including a smaller curated AI/SDG policy sub-corpus, rather than as the full universe of AI sustainability policy:

1. **Curated AI and SDG policy documents** (15,639 segments from 95 documents): This collection combines 31 systematically scraped policy documents with 64 manually curated policy documents. It includes the UN 2030 Agenda for Sustainable Development (United Nations, 2015), the Paris Agreement (UNFCCC, 2015), seven UN SDG Progress Reports (2017–2024), IPCC Sixth Assessment Report Working Group II and III Summaries for Policymakers (IPCC, 2022), the OECD AI Principles (OECD, 2019), the EU AI Act (European Union, 2024), the UNESCO Recommendation on the Ethics of AI (UNESCO, 2021), the UK National AI Strategy (UK Government, 2021), the UN AI Advisory Body Report (UN AI Advisory Body, 2024), the African Union Continental AI Strategy (African Union, 2024), SDSN (Sustainable Development Solutions Network) Sustainable Development Reports 2024 and 2025 (Sachs et al., 2024, 2025), and additional national AI strategies and international frameworks.

[1] Complete query parameters are documented in `code/0_fetch/fetch_openalex.py`. Each query filters to one OpenAlex native SDG identifier, publication year after 2017, and records with abstracts, then applies the AI search term.

2. **SDGi Voluntary National Review and Voluntary Local Review corpus** (SDGi, 2024) (31,971 segments from 313 documents): National and local review reports submitted to the UN High-Level Political Forum, covering multiple countries and reporting years. These represent state-level sustainability policy discourse.
3. **UN General Debate Corpus** (Harvard Dataverse, 2024) (6,472 segments from 2,048 documents): SDG-relevant passages filtered from UN General Debate speeches, sessions 70–80 (2015–2024), extracted by keyword filtering on SDG-related terms.

Each source document was divided into segments of approximately 150–300 words using sentence-boundary-aware dynamic segmentation. Short fragments (fewer than 20 words) were discarded. Segments were deduplicated by exact text match after normalisation, yielding the final 54,082 unique segments. Segment identifiers encode source document identity, enabling document-level weighting in subsequent analyses.

Source heterogeneity note. The three source collections represent structurally different document types. The curated AI/SDG documents are AI governance and sustainability strategy texts; VNR/VLR reports are national government self-assessments of SDG progress structured around the 17 goals; UNGDC speeches are annual head-of-state addresses that follow UN discourse conventions. SDG profiles in Section 4 should be read with this in mind: the full-corpus policy distribution reflects international institutional SDG discourse more broadly than it reflects AI governance specifically. Where an AI-governance-specific interpretation is needed, the curated AI/SDG sub-corpus (95 documents) is the more relevant reference. Appendix A.2 reports a source-family sensitivity check explicitly.

3.4 Embedding Model and Normalisation

An embedding model converts each text into a point in a 384-dimensional semantic space, such that texts with similar meaning are positioned near each other. Cosine similarity measures how close two points are in this space, ranging from 0 (no directional alignment) to 1 (identical orientation). All texts were encoded using `all-MiniLM-L6-v2` (Sentence-Transformers, 2021a), a pre-trained sentence embedding model (BERT: Bidirectional Encoder Representations from Transformers) from the sentence-transformers library (Reimers & Gurevych, 2019), producing 384-dimensional dense vector representations. This model was selected for three reasons: it achieves competitive performance on semantic textual similarity benchmarks, it is computationally efficient on CPU hardware, and, most importantly, it avoids the need for any supervised training or fine-tuning on the data under analysis. Because no model weights are adjusted and no gradient-based optimisation is applied, the embedding space is treated as a stable pre-existing coordinate system — a fixed semantic landscape rather than an object of training. This eliminates the risk of overfitting structurally rather than merely mitigating it through regularisation or cross-validation, and preserves the interpretability of every assignment as a direct geometric measurement rather than as the output of an optimised decision boundary. All embeddings were L2-normalised to unit vectors prior to storage, such that cosine similarity between any two vectors equals their dot product. This normalisation is applied consistently across all corpora and is required for the centroid computation to yield cosine similarities.

The canonical analysis uses `all-MiniLM-L6-v2`, but all key findings are replicated using `all-mpnet-base-v2` (768-dimensional) as an alternative encoder (Sentence-Transformers, 2021b). This replication is integral to the robustness framework: the cross-sensitivity comparison (Table 3) uses MPNet alongside reference-source and policy-source variations to identify which SDG-level patterns are stable enough for interpretation, and the multi-configuration H1 test (Table 2) confirms the near-zero coverage–semantic correlation under MPNet. Appendix D reports the full per-SDG comparison.

3.5 SDG Measurement Instrument: Centroids and Validation

A centroid is the average point of all text vectors assigned to a given SDG, representing that goal's typical semantic location. The core measurement instrument is a set of 17 per-SDG centroid vectors in the embedding space. Each centroid is built from all available single-label texts across four independent annotated corpora: the OSDG Community Dataset (30,534 texts, SDGs 1–16), the IISD SDG Knowledge Hub (2,221 single-label texts, SDGs 1–17), the SDGi Corpus (5,233 single-label texts, SDGs 1–17), and the Aurora dataset (5,619 expert-validated texts, SDGs 1–17). For SDGs 1–16, OSDG provides 718–3,401 texts per goal, supplemented by 256–364 SDGi texts, 18–346 Knowledge Hub texts, and 9–1,042 Aurora texts. For SDG 17, which OSDG does not cover, the centroid is built from 616 Knowledge Hub, 321 SDGi, and 557 Aurora texts (1,044 total). The SDG Classification Benchmark is held out as an independent validation set and does not contribute to any centroid. Each raw centroid vector (the mean of unit vectors) is L2-normalised to a unit vector before use, so that downstream dot products equal cosine similarities.

A new text is assigned to the SDG whose centroid is directionally nearest under cosine similarity. Every assignment is therefore traceable to a single vector comparison rather than to a learned decision boundary whose internal logic is opaque. This geometric framing is deliberate: the embedding space functions as a frozen coordinate system, and the centroid procedure measures pre-existing spatial proximity rather than constructing a classifier optimised for a specific benchmark. No model weights are fitted, no gradient descent is used, and no train/test split is required for parameter estimation; the only learned representation is the pre-trained sentence encoder. A supervised classifier trained on the same labelled data could achieve higher validation accuracy, but it would do so by carving decision boundaries around the benchmark's specific vocabulary patterns, thereby hiding the very semantic overlaps — the SDG 1/8/10 cluster, the SDG 13/17 leakage, the SDG 4 lexical artefact — that this study treats as informative findings. Because the instrument involves zero supervised training, its macro-F1 of 0.716 on the held-out benchmark (Section 4.1) reflects the intrinsic discriminability of the 17 SDGs in pre-trained semantic space — evidence that these policy concepts possess recoverable semantic signatures using only spatial proximity, not the skill of an optimised classifier. Its purpose is therefore not industry-grade operational labelling, but a reproducible semantic proxy for comparing broad corpus-level patterns.

Validation. The instrument is validated using the full SDG Classification Benchmark (616 texts, all 17 SDGs) as an independent evaluation set. For each benchmark text, the predicted SDG is the centroid with the highest cosine similarity. The evaluation metric is macro-F1 (the average per-SDG F1 score — a balanced measure of precision and recall for each category, ranging from 0 to 1) computed over all 17 SDGs (n = 616). No contamination occurs: every centroid is built from corpora independent of the benchmark (OSDG, Knowledge Hub, SDGi, Aurora). The instrument achieves macro-F1 = 0.716, comfortably above the pre-registered pass threshold of 0.50 and far above the random 17-class baseline of approximately 0.059. This level of performance would not justify deployment as a high-stakes operational SDG labeller, but it is sufficiently discriminative for the study's purpose: exploratory corpus-level comparison of broad SDG coverage and semantic framing. Per-SDG F1 scores vary: SDG 4 (Quality Education) achieves the highest F1 (0.907) and SDG 16 (Peace, Justice and Strong Institutions) the second highest (0.870), reflecting their lexically distinctive vocabularies. Notably, the very high SDG 4 F1 means the instrument is reliably assigning texts to that SDG — which strengthens rather than weakens the SDG 4 lexical-artefact concern (Section 3.9): ML papers are being confidently and consistently placed near the Education centroid. SDG 11 (Sustainable Cities, F1 = 0.545) and SDG 8 (Decent Work and Economic Growth, F1 = 0.548) are the lowest-performing, reflecting semantic overlap with SDG 9 and the SDG 1/8/10 cluster respectively. SDG 15 (Life on Land, F1 = 0.683) is also below the mean and warrants corresponding caution. SDG 17 (Partnerships, F1 = 0.300) is the most challenging goal for the instrument, consistent with its broad thematic scope. Findings for these SDGs are reported with appropriate caveats.

The validation task should also be interpreted correctly. It tests whether positive benchmark examples are assigned to the nearest appropriate SDG; it does not evaluate a rejection option for texts unrelated to any SDG. This is acceptable for the present design because the research corpus is first restricted to an OpenAlex AI-and-SDG retrieval universe. The centroid instrument then answers a relative question: among the 17 SDGs, which one is this eligible text closest to?

Additional diagnostic checks — PCA visualisations of the embedding spaces, centroid-structure diagnostics, a softmax-weighted multi-label alternative, policy source-family sensitivity, an SDG 4 lexical audit, and a directional asymmetry check — are reported in Appendix A. The canonical analysis retains the hard nearest-centroid assignment as a transparent and reproducible anchoring procedure.

3.6 Coverage Gap Analysis

For each item in the research and policy corpora, the predicted SDG is the centroid with the highest cosine similarity score (hard assignment). The coverage profile for a corpus is the proportion of items assigned to each SDG.

A critical methodological issue arises for the policy corpus: the three source collections differ dramatically in size (SDGi: 31,971 segments; curated docs: 15,639 segments; UNGDC: 6,472 segments) and the SDGi corpus includes 313 review documents, while the UNGDC material contributes 2,048 shorter speech documents. A simple segment-weighted profile would be dominated by the volume of a small number of large documents rather than reflecting the range of policy voices. I therefore compute a *document-weighted* policy coverage profile, in which each of the 2,456 unique source documents contributes equally regardless of segment count. Each document's SDG assignment is the SDG with the highest mean segment score (hard assignment). Document-weighted profiles are used as the canonical representation throughout; segment-weighted profiles are retained as a diagnostic comparison.

The coverage gap for SDG j is defined as:

$$\text{CoverageGap}_j = \left|\text{ResearchProportion}_j - \text{PolicyProportion}_j^{\text{docwt}}\right|$$

3.7 Semantic Gap Analysis

The semantic gap measures within-SDG semantic divergence: given that both corpora have texts assigned to SDG j, how differently do they discuss it? As elsewhere in this paper, "semantic" refers to distributional proximity in embedding space, not to semantic understanding in the human interpretive sense. For each SDG j:

1. Collect all research papers assigned to SDG j (the *research cluster* for j).
2. Collect all policy segments assigned to SDG j, subject to a per-document segment cap of 50 segments per source document (the *policy cluster* for j, capped). The cap prevents SDSN (up to 3,179 segments in a single document) from dominating the policy sub-centroid.
3. Compute the L2-normalised mean embedding of each cluster: $\mathbf{c}_j^{\text{res}}$ and $\mathbf{c}_j^{\text{pol}}$.
4. Define: $\text{SemanticSimilarity}_j = \mathbf{c}_j^{\text{res}} \cdot \mathbf{c}_j^{\text{pol}}$ (dot product of unit vectors = cosine similarity).

5. Define: $\text{SemanticGap}_j = 1 - \text{SemanticSimilarity}_j$.

A semantic gap of 0 would indicate perfectly overlapping semantic directions; a gap approaching 1 would indicate orthogonal framing. Robustness is assessed by replicating the analysis with segment caps of 20 and 100 per document; rankings were stable across all three caps.

The raw within-SDG centroid distance is retained as the primary measure of semantic divergence because it directly corresponds to the quantity the analysis measures: the observed difference in semantic framing between AI-for-sustainability research and SDG policy discourse within the same SDG. Additional register-adjustment procedures were explored as robustness and diagnostic checks. These analyses suggest that broad research-policy register differences may contribute to the observed gaps, but also show that more aggressive adjustment procedures risk removing substantive within-SDG differences rather than merely isolating purely superficial wording effects. Appendix E gives the projection definition explicitly and reports only register-adjusted sensitivity checks, not replacement estimates. For this reason, adjusted results are treated as sensitivity analyses rather than replacement estimates.

3.8 Coverage–Semantic Interaction

The first hypothesis (H1: coverage–semantic relationship) is that coverage gaps and semantic gaps are correlated across SDGs. I compute Pearson r (linear correlation) and Spearman ρ (rank correlation, measuring how similarly two sequences are ordered) between research coverage proportions and semantic gap scores across all 17 SDGs, and report sensitivity analyses excluding SDG 4 (see Section 3.9).

A bidirectional asymmetry diagnostic was also explored, but it is reported only in Appendix B.5 because the observed asymmetry is smaller than the systematic policy–research score gap and is therefore not treated as a headline result.

3.9 Key Assumptions and Mitigations

Corpus-level score asymmetry. Policy segments score systematically higher against the combined centroids than research papers do (mean 0.560 vs 0.218, difference of 0.342). This may reflect residual genre effects from the multi-source centroid (OSDG, IISD, SDGi, Aurora) or the substantive fact that policy documents address SDG implementation more directly — the two cannot be separated with the present design. Critically, this ambiguity does not affect the primary findings: coverage uses hard assignment rather than absolute score levels, gap rankings are the headline quantities, and both are robust to the asymmetry. Appendix A.1 reports per-source validation.

Text-unit asymmetry. Research papers are single, focused abstract-length texts (mean ∼200 words); policy corpus items are 150-word segments from documents of highly variable length. Document-weighting and per-document segment caps are the primary mitigations.

SDG 4 lexical artefact. The research coverage profile assigns 13.8% of papers to SDG 4 (Quality Education). This is flagged as artefact-affected because machine learning papers frequently use "learning," "training," and "model" as core vocabulary, which overlaps with the OSDG education corpus. Appendix A.3 shows that SDG 4-assigned texts contain education-specific vocabulary far more often than a matched non-SDG4 sample, but also frequently co-occur with ML-learning vocabulary. The result is therefore retained as a learning-related SDG 4 assignment rather than a clean education estimate. Sensitivity analyses excluding SDG 4 are provided for all correlation results.

SDG 1/8/10 overlap cluster. SDG centroids for SDGs 1 (No Poverty), 8 (Decent Work), and 10 (Reduced Inequalities) show very high pairwise cosine similarities (0.794–0.903), producing systematic classifier leakage. Individual findings for these three SDGs are reported as a macro-cluster in the discussion.

4 Results

4.1 SDG Measurement Instrument Validation

Table 1 reports per-SDG F1 scores from the benchmark evaluation (all 17 SDGs, n = 616). The instrument achieves macro-F1 = 0.716, comfortably above the PASS threshold of 0.50 set a priori. Accuracy is 0.735. The random baseline for a 17-class classifier is $1/17 \approx 0.059$; the instrument performs at 12.2 times the random baseline. Because no supervised training contributed to this score, it measures the degree to which the 17 SDGs occupy naturally distinguishable regions in a pre-trained semantic space rather than the performance of an optimised classifier.

Three SDGs require specific caveats. SDG 11 (Sustainable Cities, F1 = 0.545) is most frequently confused with SDG 9 (Innovation), whose centroid is its nearest neighbour (cosine similarity 0.720). Smart-city and urban-AI research vocabulary closely resembles innovation language, meaning SDG 11 coverage in the research corpus may be underestimated and SDG 9 coverage overestimated. SDG 8 (Decent Work, F1 = 0.548) is embedded within a high-similarity cluster with SDGs 1 and 10 (centroid similarities 0.903 and 0.794 respectively); findings for this cluster are reported with caution. SDG 15 (Life on Land, F1 = 0.683) is below the mean and may partially absorb misclassified SDG 14 items; findings for SDG 15 should be treated with corresponding uncertainty. SDG 4 achieves the highest F1 (0.907) and SDG 16 the second highest (0.870), confirming that both education/learning and governance/institutions vocabularies are sufficiently distinctive for reliable classification.

Table 1. Per-SDG F1 scores from benchmark validation across embedding models and reference sources (n=616). The benchmark is fully independent of centroid construction. Macro-F1 is the unweighted mean over available SDGs per configuration (OSDG covers 16 SDGs; all others cover 17).

	Model		Reference source			
SDG	Can	MPNet	OSDG	SDGi	KH	Aur
SDG 1	0.687	0.676	0.697	0.708	0.000	0.522
SDG 2	0.854	0.848	0.844	0.780	0.792	0.567
SDG 3	0.680	0.630	0.583	0.607	0.678	0.630
SDG 4	0.907	0.918	0.920	0.879	0.857	0.872
SDG 5	0.737	0.667	0.712	0.658	0.649	0.638
SDG 6	0.878	0.857	0.869	0.839	0.848	0.650
SDG 7	0.878	0.885	0.871	0.869	0.860	0.882
SDG 8	0.548	0.554	0.523	0.407	0.362	0.327
SDG 9	0.678	0.727	0.618	0.615	0.552	0.160
SDG 10	0.733	0.655	0.712	0.600	0.150	0.108
SDG 11	0.545	0.549	0.509	0.571	0.630	0.538
SDG 12	0.684	0.725	0.701	0.699	0.659	0.508
SDG 13	0.694	0.693	0.644	0.623	0.604	0.227
SDG 14	0.822	0.811	0.822	0.789	0.784	0.613
SDG 15	0.683	0.753	0.683	0.699	0.667	0.571
SDG 16	0.870	0.853	0.845	0.800	0.780	0.781
SDG 17	0.300	0.302	–	0.437	0.316	0.267
Macro-F1 (SDGs 1–17)	**0.716**	**0.712**	**0.722**	**0.681**	**0.599**	**0.521**

4.2 Coverage Gap: What SDGs Each Corpus Prioritises

Figure 1 presents the document-weighted policy coverage profile alongside the research coverage profile. The two profiles are markedly different.

Policy corpus. Three SDGs dominate the document-weighted profile of the full institutional policy corpus: SDG 13 (Climate Action, 25.0%), SDG 17 (Partnerships for the Goals, 48.2%), and SDG 16 (Peace, Justice and Strong Institutions, 14.5%), accounting for 87.7% of policy documents. This is a sharply policy-facing governance cluster: climate commitments, international coordination, and institutional capacity dominate the policy surface much more than sector-specific programme areas. Appendix A.2 shows that this full-corpus profile is source-family sensitive: the curated AI/SDG sub-corpus is much more innovation-heavy, while the broader SDGi and UNGDC components are more climate-, partnerships-, and governance-oriented.

Research corpus. The research profile is broader, but it still concentrates strongly on a smaller set of SDGs. The largest research shares are SDG 3 (Good Health, 17.9%), SDG 4 (Quality Education; flagged as artefact-affected, 13.8%), and SDG 9 (Innovation, 10.2%), followed by SDG 16 and SDG 7 at 8.4% and 8.4% respectively. Appendix A.3 reports a lexical audit supporting the SDG 4 caveat. Even with that caution, the research corpus clearly does not mirror the policy-side dominance of SDGs 13, 17, and 16.

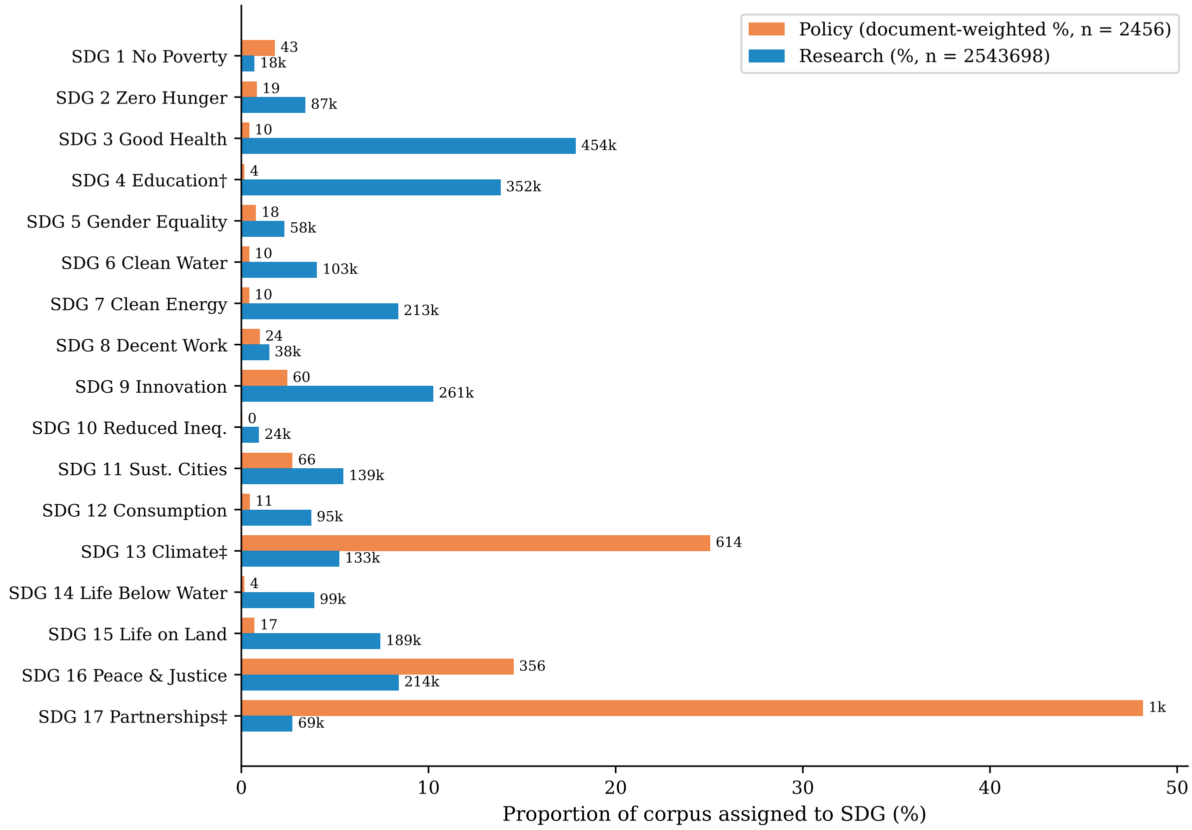

Figure 1. Coverage profiles by SDG: proportion of each corpus assigned to each SDG via hard assignment (highest-scoring SDG). Policy proportions are document-weighted (n = 2,456 source documents); research proportions are paper-weighted (n = 2,543,698 papers). [(a)]SDG 4 research share likely inflated by ML vocabulary artefact (see Section 3.9); its F1 = 0.907 in Table 1 indicates confident assignment, which strengthens the artefact concern. [(b)]SDG 13 and SDG 17 centroids are moderately to strongly correlated (cosine similarity = 0.805); combined share = 73.2%. Total coverage gap across all SDGs = 1.450 (mean per-SDG = 0.085).

Largest coverage mismatches. The five largest absolute coverage gaps are SDG 17 (0.454), SDG 13 (0.198), SDG 3 (0.175), SDG 4 (0.137), and SDG 7 (0.080). This produces a clear structural contrast: international institutional SDG policy discourse is heavily concentrated on the climate–partnerships–governance block, while research attention is pulled much more toward health, learning-related material, and innovation.

SDG 13/17 collinearity note. The SDG 13 and SDG 17 centroids have a pairwise cosine similarity of 0.805, comparable in magnitude to pairwise similarities within the SDG 1/8/10 cluster (0.794–0.903). Climate discourse and partnership discourse are semantically intertwined in the combined reference corpora, producing substantial leakage between these two classifications. The individual figures of 25.0% (SDG 13) and 48.2% (SDG 17) are each subject to this leakage and should be treated with the same caution as the SDG 1/8/10 cluster. The combined SDG 13+17 policy share (73.2%) is the more interpretively stable figure. In both cases, the core finding — that this cluster dominates the policy profile while research concentrates elsewhere — is robust.

Score note. The mean top-centroid score for policy segments (0.560) exceeds that for research papers (0.218) by 0.342, above the pre-registered flag threshold of 0.10. This gap is consistent with either residual genre effects in the combined centroid or the substantive interpretation that policy documents address SDG implementation more directly. Coverage proportions are based on hard assignment (which SDG scores highest) rather than absolute score levels, so the ambiguity does not affect the primary findings.

4.3 Semantic Gap: How Differently Research and Policy Discuss Each SDG

Figure 2 shows the within-SDG semantic gap for all 17 SDGs. Semantic gaps range from 0.233 (SDG 9) to 0.658 (SDG 17), a span of 0.424 cosine distance units. Rankings are stable across segment caps of 20, 50, and 100, confirming robustness to the document sampling parameter.

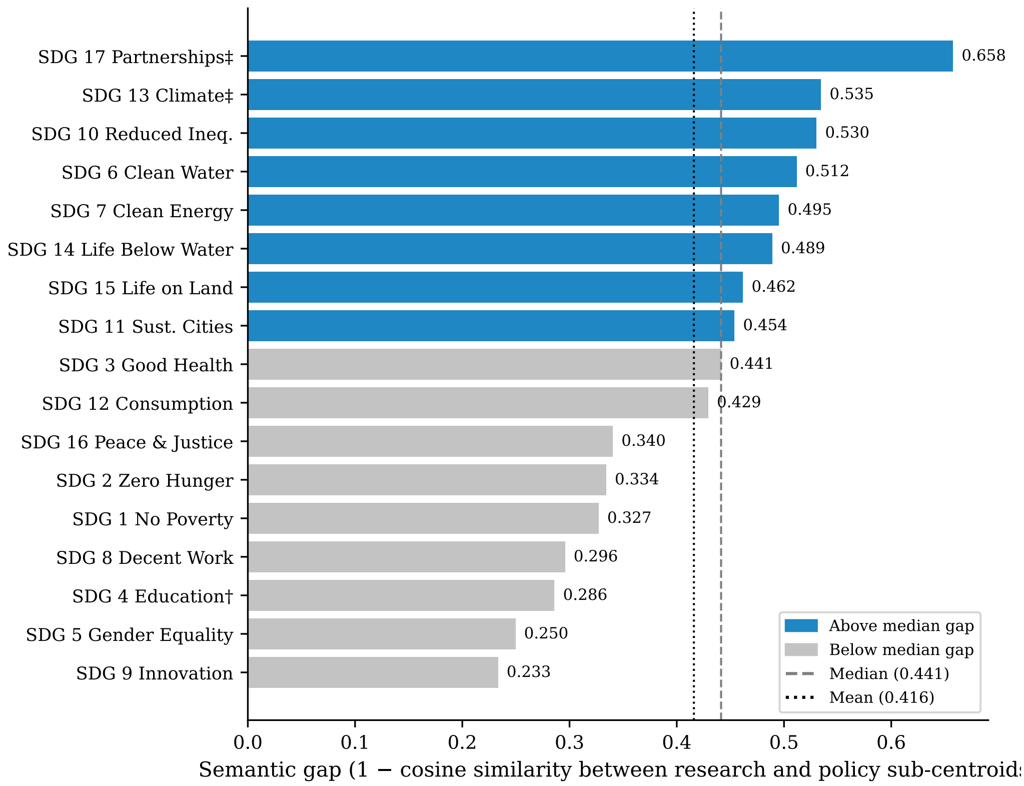

Figure 2. Within-SDG semantic gap by SDG, sorted descending. Semantic gap = 1− cosine similarity between the research sub-centroid and the policy sub-centroid for each SDG (segment cap = 50 per source document). Dashed line = median (0.441); dotted line = mean (0.416). Dark bars exceed the median; light bars fall below it.

Largest semantic gaps. The greatest within-SDG divergence appears in SDG 17 (Partnerships, 0.658), SDG 13 (Climate Action, 0.535), and SDG 10 (Reduced Inequalities, 0.530), with SDGs 6 and 7 close behind. This pattern is robust across reference sources: single-source centroids produce similar validation F1 scores to the combined centroid for most SDGs, with OSDG-only and combined F1 agreeing within 0.05 for all 16 available SDGs (Appendix A.1). Notable exceptions are SDG 10 (Reduced Inequalities), where the Knowledge Hub centroid produces F1 = 0.150 (vs combined 0.733), and SDG 8 (Decent Work), where both SDGi-only and KH-only centroids underperform the combined (0.407 and 0.362 vs 0.548), indicating that the supplementary sources broaden the semantic coverage of narrower OSDG-only clusters. The combined centroid now also incorporates the Aurora expert-validated research corpus (Vanderfeesten et al., 2020) (5,619 texts, all 17 SDGs), adding a research-domain perspective alongside the OSDG, SDGi, and Knowledge Hub sources. Appendix A.1 reports per-source validation for all five sources. The substantive point remains the same: some of the most institutionally salient policy goals are also among the most semantically distant. Where institutional policy discourse is most concentrated, research does not just appear less often; it also tends to frame those SDGs differently when it does appear.

Smallest semantic gaps. The lowest gaps occur in SDG 9 (0.233), SDG 5 (0.250), SDG 4 (0.286), and SDG 8 (0.296). Low semantic gap does not imply balanced attention: SDG 9 is still strongly research-dominant in coverage, and SDG 4 remains subject to the learning-vocabulary caveat discussed in Appendix A.3. The key point is narrower: some SDGs show relatively close within-goal framing even when the two corpora differ sharply in how much attention they allocate to that goal. Appendix B.3 gives a compact lexical interpretability check for SDGs 17, 13, and 9 to show what these high- and lower-gap cases look like in text.

4.4 Coverage–Semantic Interaction: Two Distinct Dimensions of Observed Divergence

Table 2 reports the H1 coverage–semantic correlation tests between coverage measures and semantic gap scores across the SDGs.

The primary H1 test is **not supported**. Research coverage proportion and semantic gap are almost uncorrelated across SDGs ($r = -0.161$, $p = 0.536$; $n = 17$). The 95% confidence interval (the range plausibly containing the true value) for r spans approximately $[-0.60, +0.35]$, so the result should be read as low-evidence for a linear relationship rather than proof of full independence.

The diagnostic scatter plot (Figure 3) provides a per-SDG complement to the aggregate correlation test. SDGs with larger absolute coverage gaps tend to have larger semantic gaps, and the most policy-dominant SDGs tend to sit above the median on both axes. In other words, sheer research attention does not predict semantic distance, while strong cross-corpus imbalance shows a descriptive tendency to coincide with larger semantic gaps in this dataset.

Table 2. H1 correlation tests (Pearson r, Spearman ρ) between research proportion and within-SDG semantic gap. The primary test uses all 17 observed SDGs; the sensitivity test excludes SDG 4. Additional rows replicate the test under an alternative embedding model (MPNet) and four alternative reference-source centroids (OSDG, SDGi, Knowledge Hub, Aurora), all showing near-zero correlations consistent with the canonical result.

Test	Pearson r	p	Spearman ρ	p
Primary observed SDGs (n = 17)	-0.161	0.536	-0.093	0.722
Excluding SDG 4	-0.044	0.873	0.026	0.922
MPNet model (n=17)	-0.084	0.749	0.049	0.852
OSDG-only centroids (n=16)	-0.030	0.913	0.047	0.863
SDGi-only centroids (n=17)	0.122	0.642	0.176	0.498
Knowledge Hub-only centroids (n=17)	0.023	0.929	0.164	0.529
Aurora-only centroids (n=17)	-0.057	0.829	0.127	0.626

Figure 3 illustrates this diagnostic refinement. High-coverage research goals are split between relatively low-gap cases such as SDG 9 and higher-gap cases such as SDG 3 and SDG 7, which is why the primary H1 test is near zero. At the same time, the most policy-dominant goals, especially SDGs 13 and 17, sit above the median on both axes. As a descriptive observation, the substantive lesson is therefore not that coverage and semantic divergence are unrelated in every sense, but that *research attention alone* is a poor predictor of semantic convergence.

4.5 Cross-Sensitivity Robustness of the Gap Rankings

The semantic gap rankings discussed below are not equally stable across all measurement choices. Table 3 compares each SDG's gap rank under the canonical configuration (MiniLM encoder, combined centroids, full policy corpus) against three families of alternatives: an alternative embedding model (MPNet), five reference-source configurations (OSDG-only, SDGi-only, Knowledge Hub-only, Aurora-only), and four policy sub-corpora (curated AI/SDG, SDGi VNR/VLR, UNGDC). Each cell reports the within-SDG gap rank where 1 = largest gap.

Three tiers of robustness emerge directly from the data. First, a small set of SDGs maintain their relative positions across all or nearly all configurations. SDG 17 is the top-gap SDG in the canonical analysis and remains among the top two under model substitution and all policy-family checks, though its rank varies considerably under alternative reference sources (rank 8 under SDGi-only centroids, rank 16–17 under Knowledge Hub and Aurora). SDGs 9 and 5 are consistently among the lowest-gap SDGs across all measurement configurations, typically occupying the bottom three ranks. Second, a cluster of SDGs (6, 7, 13, 14) consistently occupy the upper half of the gap ranking across all checks, though their exact ordering shifts by no more than three positions. Third, the remaining SDGs (1, 2, 3, 8,

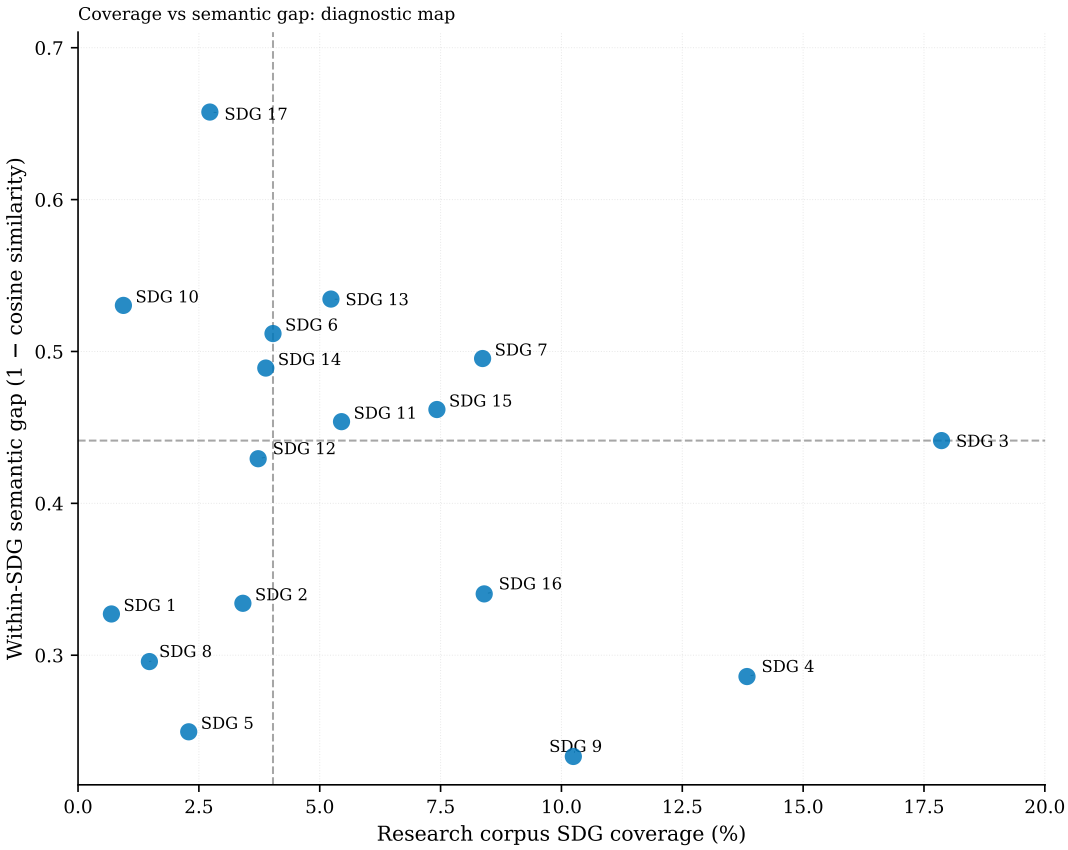

Figure 3. Coverage vs semantic gap scatter plot (diagnostic map). Each point is one SDG. Reference lines indicate the median research coverage (4.03%) and median semantic gap (0.441). The figure shows that high research attention can coexist with either low or high semantic divergence, while the most policy-dominant SDGs tend to sit above the median on both axes.

10, 11, 12, 15, 16) are sensitive to measurement choices: their ranks shift by four or more positions under model substitution alone, with additional variation across reference and policy sources. The discussion below limits its interpretive claims to findings that survive this filtering.

5 Discussion

5.1 Two Distinct Dimensions of Observed Divergence

The cross-sensitivity comparison (Table 3) establishes empirical bounds on which gap findings are stable enough to sustain interpretation. The discussion below is limited to patterns that survive these checks. The headline result of this study is simple: shared SDG labels are insufficient evidence of shared framing. This operationalises the Level 1 to Level 2 advance: where earlier bibliometric work could report only that SDG 9 attracts more research attention than SDG 17, embedding-based comparison additionally shows that SDG 17's research-policy framing is substantially more divergent than SDG 9's. The rebuilt

Table 3. Cross-sensitivity robustness of within-SDG semantic gap rankings. Each cell reports the gap rank (1 = largest gap, 17 = smallest gap) under each measurement configuration. A "–" indicates that the source does not cover that SDG.

	Model		Reference source					Policy source			
SDG	Can	MPNet	Combined	OSDG	SDGi	KH	Aur	Full	AI/SDG	SDGi-VNR	UNGDC
SDG 1	13	7	15	12	17	1	8	13	11	13	13
SDG 2	12	11	11	11	10	7	6	12	10	14	12
SDG 3	9	4	10	8	7	8	9	9	12	7	10
SDG 4	15	14	13	13	16	13	12	15	15	16	15
SDG 5	16	17	16	16	14	15	15	16	16	17	17
SDG 6	4	6	4	3	5	4	1	4	2	4	6
SDG 7	5	1	3	4	3	2	7	5	3	6	4
SDG 8	14	16	14	14	11	9	16	14	13	15	11
SDG 9	17	15	17	15	13	17	14	17	17	12	16
SDG 10	3	10	8	2	6	10	11	3	5	2	3
SDG 11	8	3	7	7	9	11	5	8	9	8	2
SDG 12	10	8	9	9	12	12	2	10	7	10	5
SDG 13	2	9	2	1	2	5	3	2	6	3	9
SDG 14	6	5	5	5	1	3	4	6	8	5	7
SDG 15	7	12	6	6	4	6	10	7	4	9	8
SDG 16	11	13	12	10	15	14	13	11	14	11	14
SDG 17	1	2	1	–	8	16	17	1	1	1	1

canonical analysis shows that research attention alone is a poor predictor of semantic convergence. SDGs that attract substantial research attention can still be either relatively close to policy framing (for example, SDG 9) or substantially divergent from it (for example, SDGs 3 and 7). At the same time, the strongest policy-dominant SDGs, especially SDGs 13 and 17, are also among the most semantically distant. The sample-stability ladder (Appendix C) strengthens this reading: once the research subsamples reach roughly 200,000 papers, the headline downstream metrics already sit very close to the full-corpus values. The practical implication is that observed research–policy divergence is not a single scalar problem. Attention allocation and framing divergence must be measured separately.

This pattern is best interpreted as a structural difference between knowledge systems rather than as a simple failure of communication. The Mode 1/2 distinction (Gibbons et al., 1994) sharpens this reading. The research corpus concentrates on technically tractable applications in health, energy, and infrastructure — SDGs 3, 7, and 9 — where dominant ML tasks align with well-defined problem-solving rather than institutional coordination, while the policy-side dominance of SDGs 13, 16, and 17 reflects Mode 2 goal-setting arising from contexts of application that demand coordination and collective action. Epistemic-community theory helps clarify the implication without overstating it: if expert influence partly depends on shared problem framing under uncertainty (Haas, 1992), then large within-SDG semantic gaps indicate where such shared framing may be less plausible. Following Cross (2013), however, semantic proximity is only a possible precondition for knowledge transfer; it cannot show whether expert communities exist, whether they have policy access, or whether policy actors use the relevant research. The two-communities theory (Caplan, 1979) reinforces this structural reading: the uncorrelatedness of coverage and semantic gap is what one would expect if the two corpora represent distinct knowledge cultures with different reward systems and knowledge-use criteria, rather than two sides of a shared enterprise.

This two-dimensional pattern is also robust to the most aggressive controls for register differences between research and policy text varieties. Register-adjustment sensitivity checks (Appendix E) that learn and subtract the broad research-policy embedding direction reduce semantic gaps but do not eliminate the underlying ranking. Even the most aggressive within-SDG adjustment — which risks over-subtracting the substantive contrast of interest — compresses but preserves the relative ordering of high-gap and low-gap SDGs. The spread between raw and adjusted estimates therefore provides a plausible uncertainty band around the register-sensitive component of the gap, but the finding that some SDGs show relatively close framing while others show substantial divergence is not an artefact of broad genre differences in writing style.

More broadly, this work suggests that research–policy alignment is not a single measurable property but a hierarchy of increasingly demanding forms of correspondence, of which semantic proximity is only one intermediate level. What existing bibliometric and keyword-based studies capture as "alignment" is predominantly lexical or topical correspondence (Level 1); the present study extends measurement to semantic proximity (Level 2); but full substantive alignment — shared problem definitions, implied interventions, and implementation pathways (Level 3) — remains beyond the reach of unsupervised text analysis and would require structured qualitative or mixed-methods follow-up. The implication is not that lower-level measures are uninformative but that they need to be explicitly situated within this hierarchy rather than treated as proxies for alignment as a whole.

The measurement instrument that produces these patterns is deliberately zero-shot: it involves no supervised training, no gradient-based optimisation, and no adjustment of the embedding space to the data under analysis. The centroid of each SDG is the geometric mean of its labelled reference texts, and every document assignment is the result of a single cosine comparison to that centre of mass. This design choice is consequential for how the validation and gap results should be interpreted. A supervised classifier trained on the same labelled benchmark could — through aggressive regularisation or a larger training set — achieve a higher macro-F1 than the 0.716 reported here. But such a classifier would learn to draw boundaries around the benchmark's specific vocabulary and annotation conventions, thereby transforming the task from semantic measurement to predictive optimisation. The overlaps that the present instrument exposes — the SDG 1/8/10 centroid cluster, the SDG 13/17 leakage, the SDG 4 learning-vocabulary artefact — would not disappear under a supervised model; they would simply be hidden inside a learned decision function that forced hard separations where the underlying language is not cleanly separated. Because this study asks a diagnostic question — where do research and policy discourse diverge when measured through a transparent geometric proxy — rather than an operational one — can a text be assigned to its correct SDG with minimal error — the zero-shot centroid approach is fit for purpose. It provides evidence that the 17 SDGs possess recoverable semantic structure in a general-purpose language space without any data-specific training, and it preserves the overlaps and ambiguities as findings rather than engineering them away.

5.2 Robust Patterns of Divergence

The cross-sensitivity comparison narrows the set of gap findings that are stable enough to sustain individual interpretation. Only two SDGs maintain their extreme positions across all or nearly all measurement configurations: SDG 17 as the robustly highest-gap SDG, and SDG 9 as the robustly lowest-gap SDG. SDG 5 is also consistently among the lowest-gap goals across all checks.

SDG 17's top-gap position is stable under model substitution (rank 1 under MiniLM, rank 2 under MPNet), under all four policy sub-corpora (rank 1 in every family), and under OSDG-based centroids. Under alternative reference sources (SDGi-only, Knowledge Hub-only, Aurora-only), however, its rank varies considerably, reflecting the sensitivity of the combined centroid to source composition for this SDG. The claim that SDG 17 is the most divergent goal is therefore conditional on the combined reference source rather than instrument-independent.

This consistently largest gap is consistent with SDG 17's distinctive structural role. Le Blanc (2015) characterises SDG 17 as the dedicated goal for cross-cutting means of implementation — encompassing finance, technology transfer, capacity-building, and partnerships — distinct from the substantive domains of the other 16 goals. The Griggs et al. (2017) guide to SDG interactions similarly describes SDG 17 as providing "the main enablers for implementing the entire SDG framework." Unlike every other SDG, where research and policy discourse both centre on a shared substantive area, SDG 17 policy discourse concerns the enabling infrastructure for other goals — partnerships, finance, institutional coordination — a conceptual space that research abstracts on technical AI applications do not typically inhabit. The gap's stability across measurement conditions may therefore partly reflect this asymmetry between what the goal is (implementation enabler) and what the research corpus reports (technical substance), rather than divergent framing of shared content alone.

SDG 9 is the robustly closest SDG, occupying the bottom two or three ranks under every measurement configuration. This does not prove substantive alignment, but it demonstrates that the embedding-based instrument can separate a relatively close case from more visibly divergent ones.

The main coverage patterns are themselves robust. The full policy corpus is overwhelmingly dominated by the climate–partnerships–governance block (SDGs 13, 17, and 16). Appendix A.2 notes that this is source-family sensitive — the curated AI/SDG sub-corpus is more innovation-oriented — but the broader institutional profile holds across the UNGDC and SDGi families. The research corpus, conversely, concentrates on technically tractable SDGs (3, 7, 9, 15, 16).

A cluster of SDGs (6, 7, 13, 14) consistently occupy the upper half of the gap ranking across all checks, though their exact ordering shifts by no more than three positions. These SDGs are reliably above the median in research–policy divergence, but their precise rank is measurement-dependent and should not be over-interpreted individually. Similarly, the within-cluster spread observed between SDG 8 and SDG 10 within the 1/8/10 centroid cluster is a structural finding that does not depend on either SDG's precise rank: it undermines the concern that gap rankings are purely a byproduct of centroid similarity, because the instrument separates cluster members even where classifier leakage is strongest.

The remaining SDGs (1, 2, 3, 8, 10, 11, 12, 15, 16) shift position by four or more ranks under model substitution alone, with additional variation across reference and policy sources. No individual interpretive claims are made about their gap positions.

5.3 The Coverage–Gap Dissociation Is Robust

The most robust aggregate finding is not about any single SDG: it is that research attention and semantic divergence are dissociated. The primary H1 test shows that coverage proportion and semantic gap are nearly uncorrelated ($r = -0.161$, $p = 0.536$; $n = 17$), and this result is insensitive to excluding SDG 4 as a suspected lexical artefact. The sample-stability ladder (Appendix C) confirms that the headline metrics are not artefacts of corpus size: once the research subsamples reach roughly 200,000 papers, the mean semantic gap, policy-to-research asymmetry, and score gap stabilise to within 0.001 of the full-corpus values. Register-adjustment checks (Appendix E) further confirm that the relative ordering of high-gap and low-gap SDGs is preserved even under the most aggressive controls for genre differences between research and policy text varieties.

The practical implication is that observed research–policy divergence is not a single scalar problem. Attention allocation and framing divergence must be measured separately.

Beyond the empirical findings, the measurement framework developed here is transferable beyond AI and the SDGs. The same approach — embedding-based centroids built from labelled reference texts, with coverage and semantic gap computed against a comparison corpus — could be applied to climate adaptation, public health, biodiversity policy, or education, wherever a reference taxonomy and comparable research and policy-facing corpora exist. The framework does not require supervised training, language-model fine-tuning, or domain-specific annotation; it requires only a frozen embedding space and labelled exemplars to define the category geometry. This generality is intentional: the contribution is a diagnostic methodology for measuring research-policy semantic proximity, not only a specific set of SDG gap numbers.

5.4 Implications

Two implications follow from the combined findings, and each pushes against the common habit of treating topical mention as evidence of alignment.

Research and policy agenda-setting. The results identify candidates for qualitative audit, policy-facing synthesis, and boundary-spanning work that examines where research and policy differ in problem definitions, evidence standards, and intended users. Rather than implying that funding should be mechanically redirected on the basis of embedding distances, the map indicates where research-policy translation, agenda-setting, and knowledge-brokerage work may be most needed. In Guston's terms, high-gap SDGs may be candidates for boundary-organising practices: forums, roles, or institutions that create usable boundary objects while remaining accountable to both knowledge producers and policy users (Guston, 2001).

SDG-aware research communication. The finding that research attention alone does not predict semantic alignment suggests that simply increasing research output on neglected SDGs is insufficient; research communication norms may matter as well. Future work could test whether publication incentives that encourage explicit SDG framing improve semantic proximity to policy discourse more effectively than topic-level funding mandates alone. In practice, this could mean funder-required policy-facing summaries, journal-level SDG tagging with short justification fields, or synthesis briefs written for policy audiences rather than only academic abstracts. Following Cash et al. (2003), this should be understood as boundary work rather than communication alone. High-gap SDGs may require not only clearer policy-facing summaries, but also translation between technical and institutional vocabularies and mediation between different standards of salience, credibility, and legitimacy. Guston (2001) sharpens this implication institutionally: effective boundary work may require dual accountability to both scientific and policy principals, rather than one-way dissemination from research to policy.

A bidirectional asymmetry diagnostic was also attempted, but because the observed asymmetry (0.152) is smaller than the systematic policy–research score gap (0.342), it is reported only in Appendix B.5 as an inconclusive sensitivity check rather than as a substantive result.

These implications speak to distinct audiences. For SDG monitoring bodies and bibliometric observatories, the finding that shared labels do not imply shared framing is a direct challenge to keyword-based SDG mapping as a proxy for alignment: coverage and framing should be reported as separate dimensions rather than collapsed into a single indicator. For research funders who evaluate grant portfolios by SDG relevance, the results suggest that funding allocation alone is unlikely to close the semantic gap without complementary attention to how research communicates its policy relevance. For journals adopting SDG classification tags, the finding that framing divergence varies substantially by SDG — and can coexist with high research output — recommends caution against treating SDG tagging as sufficient evidence of policy-aligned research.

5.5 Limitations

The cross-sensitivity comparison (Table 3) bounds which gap findings are stable enough to sustain interpretation. The limitations below address scope conditions that this comparison does not resolve.

Scope: semantic proximity, not substantive alignment or policy influence. The study measures semantic co-location in embedding space, not whether research and policy are substantively addressing the same problems or whether research is salient, credible, or used by policy actors. Embedding proximity reflects shared vocabulary, not shared agenda, and does not measure policy influence, institutional mechanisms, or knowledge use — these would require interviews, process tracing, or institutional analysis beyond this study's scope.

SDG 17 combined-source fragility. SDG 17 (Partnerships for the Goals) is the analysis's weakest measurement point. Its validation F1 is low across all centroid sources (0.300 combined, 0.437 SDGi-only, 0.316 Knowledge Hub, 0.267 Aurora), indicating the goal's broad mandate resists compact semantic definition rather than reflecting any single-source artefact. Additionally, because the combined centroid includes SDGi policy documents that share genre with part of the scored policy corpus, the within-genre overlap may marginally reduce the SDG 17 semantic gap. The cross-sensitivity table (Table 3) confirms the concern: SDG 17 is rank 1 under the combined centroid but falls to rank 8 under SDGi-only centroids. Per-source F1 and centroid cosine validation is in Appendix A.1 (Table 4).

Corpus scope and retrieval boundaries. The policy corpus represents international institutional SDG discourse (UN, OECD, national governments) rather than the full universe of AI sustainability policy. Domestic implementation plans, sub-national policies, and non-English-language documents are not represented. The research corpus is retrieved through OpenAlex's native SDG filters and is also English-only. Both biases likely understate the perspectives of the Global South. SDG-related research can also be delineated through other strategies (citation expansion, keyword ontologies, platform-assigned labels, or curated datasets), so results should be read as conditional on these corpus and retrieval boundaries. Appendix A.2 makes the policy source-family sensitivity explicit.

Corpus asymmetry: AI-for-SDG scope. The research corpus is a tight intersection of AI and SDG content (AI $\cap$ SDG), while the policy corpus is closer to a union covering both institutional SDG discourse and AI governance documents with no requirement that documents address both topics. The measured semantic gap may therefore reflect this scope mismatch — policy documents discuss sustainability topics and AI governance issues that the research corpus was never designed to contain. This does not invalidate the coverage and gap findings where the corpora overlap, but the between-SDG rankings should be interpreted as conditional on this asymmetric construction rather than as a domain-independent characterisation of AI-for-SDG alignment.

Register effects on the semantic gap. Semantic distance may capture differences in communicative function (see Biber and Conrad (2009) on the register–genre–style distinction). Research abstracts and policy documents are text varieties designed to do different things, so the semantic gap should be read as observed textual-semantic divergence, not as a purified estimate of substantive disagreement. Appendix E reports register-adjusted estimates alongside raw values; the spread between raw and aggressively adjusted estimates is best read as an uncertainty band around the register-sensitive component (cf. Biber (1988)'s argument that register differences must be investigated empirically rather than assumed from text-type categories). The raw gap is retained because it best matches what the study measures: observed research-policy semantic distance within each SDG.

Cross-sectional design. The study provides a snapshot of the 2018–2025 research period against a policy corpus assembled over a similar timeframe; it cannot address whether alignment is improving or deteriorating, or whether specific policy events (e.g., the EU AI Act 2024, IPCC AR6 2022) have shifted research framing.

6 Conclusion

This paper revisits an old problem with a new instrument. Policy scholars have long argued that research and policy often fail to meet one another on equal terms. What recent semantic tools add is not a final verdict, but the ability to map one important dimension of it directly, across thousands of texts.

The core finding is that shared SDG labels are insufficient evidence of shared framing. Research concentrates on SDGs 3, 4 (flagged as artefact-affected), and 9, while institutional policy discourse is dominated by SDGs 13, 16, and 17; within the same SDG, semantic divergence is largest in some of the most policy-dominant goals. The resulting map therefore separates cases where apparent alignment is strongest (SDG 9) from cases where shared labels conceal substantial divergence (SDGs 13 and 17). This contrast is the dissertation's main conceptual contribution: embedding-based distance can help identify where closer investigation is warranted, not as a final classification of whether any given gap is productive or problematic.

Extending the analysis toward finer SDG-target-level interpretation could build on the Aurora dataset (Vanderfeesten et al., 2020) by applying target-level rather than goal-level annotation. Disaggregating the policy corpus by country income level would provide finer-grained evidence about where observed divergence is most acute and for whom. A temporal extension — tracking whether semantic gaps narrow or widen over successive policy cycles — would test whether policy events such as the EU AI Act or IPCC assessment reports shift research framing. Finally, moving from goal-level to target-level comparison would assess whether apparent proximity at the SDG level conceals divergence at the more operationally relevant target level.

The central implication is not that the AI-for-sustainability field is aligned or misaligned in aggregate — it is both, depending on which SDG and which dimension one examines. Coverage and framing are separate axes of divergence; a field that appears well-aligned on one axis may be substantially divergent on the other. Semantic mapping cannot replace qualitative understanding of why particular gaps persist or what they mean for policy uptake, but it can show — at scale and with transparent assumptions — where the questions are worth asking.

Several methodological extensions follow naturally from this baseline. First, the nearest-centroid rule assumes uniform spherical structure around each SDG centroid; future work could learn per-SDG density-based or variance-weighted decision boundaries that better reflect each goal's semantic breadth. Second, the instrument currently assigns every text to the nearest SDG even when the closest centroid is still distant, which is acceptable for a corpus-level diagnostic but would be a limitation for screening or filtering tasks. An LLM-generated synthetic corpus of explicitly non-SDG text — sampled from domains such as corporate finance, popular culture, or pure mathematics — could serve as negative anchors, enabling out-of-distribution detection before centroid assignment. Third, a supervised classifier trained on the same benchmark data could push macro-F1 beyond the 0.716 reported here, albeit at the cost of accepting the overfitting risk that the present design structurally avoids.

These directions are not gaps in the study's execution; they represent distinct methodological battlefields. The objective of this dissertation was not to maximise a statistical metric through optimisation, but to establish whether the 17 SDGs possess recoverable semantic structure under a transparent, zero-shot geometric measurement. Proving that they do — using nothing but spatial proximity in a frozen embedding space — is the prerequisite for any subsequent tightening of class boundaries, filtering logic, or supervised refinement, not a substitute for it. Complexity should not be introduced where fundamental language geometry already yields a robust signal. If future work produces better boundaries, sharper filters, or higher F1 scores, those advances will rest on the foundation of knowing that the semantic structure exists to be optimised in the first place.

More broadly, this dissertation argues that alignment should be understood as a multi-level empirical construct — from lexical overlap through semantic proximity to substantive agreement — rather than as a binary property inferred from shared labels.

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A Diagnostics for Centroid Construction

A.1 Per-SDG Reference-Source Comparison

The canonical centroids combine OSDG, SDGi, and Knowledge Hub texts per SDG. This appendix compares each single-source centroid to the canonical combined centroid across four metrics: validation F1, centroid cosine similarity, research coverage (papers assigned per SDG), and within-SDG semantic gap. The purpose is to assess whether any single source dominates the combined centroid and whether the study's headline results are sensitive to the choice of reference corpus.

Table 4 reports per-SDG validation F1 and centroid similarity for all four configurations:

- **Combined** (canonical): all available single-label texts (OSDG + SDGi + Knowledge Hub) per SDG.
- **OSDG-only**: the OSDG Community Dataset only (SDGs 1–16; SDG 17 is not available in OSDG).
- **SDGi-only**: the SDGi Corpus only (VNR/VLR policy reports).
- **Knowledge Hub-only**: the IISD SDG Knowledge Hub only (journalism).

Table 4. Per-SDG source comparison: validation F1 and centroid cosine similarity to the canonical combined centroid. A "–" indicates no texts for that SDG in the source (OSDG, SDG 17) or a trivial self-similarity of 1.0.

SDG	Combined		OSDG		SDGi		Knowledge Hub		Aurora	
	F1	cos	F1	cos	F1	cos	F1	cos	F1	cos
SDG 1	0.687	1.000	0.697	0.990	0.708	0.914	0.000	0.512	0.522	0.823
SDG 2	0.854	1.000	0.844	0.985	0.780	0.898	0.792	0.810	0.567	0.800
SDG 3	0.680	1.000	0.583	0.959	0.607	0.872	0.678	0.732	0.630	0.879
SDG 4	0.907	1.000	0.920	0.993	0.879	0.897	0.857	0.703	0.872	0.798
SDG 5	0.737	1.000	0.712	0.996	0.658	0.890	0.649	0.725	0.638	0.843
SDG 6	0.878	1.000	0.869	0.993	0.839	0.898	0.848	0.845	0.650	0.785
SDG 7	0.878	1.000	0.871	0.995	0.869	0.906	0.860	0.847	0.882	0.904
SDG 8	0.548	1.000	0.523	0.983	0.407	0.893	0.362	0.748	0.327	0.818
SDG 9	0.678	1.000	0.618	0.992	0.615	0.850	0.552	0.730	0.160	0.757
SDG 10	0.733	1.000	0.712	0.985	0.600	0.874	0.150	0.525	0.108	0.618
SDG 11	0.545	1.000	0.509	0.991	0.571	0.898	0.630	0.749	0.538	0.892
SDG 12	0.684	1.000	0.701	0.981	0.699	0.946	0.659	0.877	0.508	0.869
SDG 13	0.694	1.000	0.644	0.986	0.623	0.915	0.604	0.900	0.227	0.774
SDG 14	0.822	1.000	0.822	0.982	0.789	0.917	0.784	0.836	0.613	0.800
SDG 15	0.683	1.000	0.683	0.992	0.699	0.933	0.667	0.861	0.571	0.875
SDG 16	0.870	1.000	0.845	0.995	0.800	0.796	0.780	0.691	0.781	0.927
SDG 17	0.300	1.000	–	–	0.437	0.937	0.316	0.982	0.267	0.821

Several patterns emerge from Table 4. First, OSDG-only centroids are very close to the combined direction for all 16 available SDGs (cosine 0.985–0.998), and their validation

Table 5. Per-SDG source comparison: research coverage (papers assigned) and within-SDG semantic gap for each reference source. Coverage is the count of research papers assigned to each SDG via nearest-centroid matching. Gap is $1 - \cos(\text{research centroid}, \text{policy centroid})$; larger values indicate greater research–policy divergence.

SDG	Combined		OSDG		SDGi		Knowledge Hub		Aurora	
	n	gap	n	gap	n	gap	n	gap	n	gap
SDG 1	11,804	0.253	18,348	0.329	14,782	0.266	35,146	0.805	21,489	0.447
SDG 2	83,441	0.324	88,364	0.334	100,220	0.344	160,806	0.449	120,798	0.505
SDG 3	470,627	0.423	461,638	0.442	421,276	0.413	425,281	0.436	377,884	0.434
SDG 4	355,720	0.287	353,269	0.287	367,183	0.289	382,814	0.298	415,692	0.317
SDG 5	53,741	0.249	58,680	0.254	81,372	0.297	71,840	0.275	49,479	0.257
SDG 6	90,715	0.491	104,730	0.514	72,936	0.430	129,462	0.522	119,951	0.595
SDG 7	198,743	0.497	216,430	0.503	233,924	0.512	340,638	0.568	150,021	0.497
SDG 8	41,898	0.269	38,662	0.286	63,828	0.317	124,609	0.379	88,471	0.241
SDG 9	247,370	0.234	271,741	0.261	240,883	0.314	131,436	0.236	297,119	0.288
SDG 10	20,323	0.432	25,249	0.516	42,974	0.429	76,100	0.372	85,597	0.328
SDG 11	136,616	0.447	141,142	0.466	91,158	0.346	108,151	0.368	190,345	0.525
SDG 12	90,960	0.424	96,433	0.430	62,947	0.316	97,454	0.355	66,345	0.577
SDG 13	110,645	0.528	155,820	0.572	141,882	0.529	62,480	0.495	160,148	0.551
SDG 14	99,162	0.485	101,402	0.492	123,624	0.534	115,658	0.527	100,242	0.545
SDG 15	175,447	0.465	194,906	0.472	214,340	0.504	186,868	0.485	107,125	0.427
SDG 16	189,046	0.323	216,884	0.368	174,806	0.297	69,880	0.289	133,242	0.317
SDG 17	167,440	0.671	0	–	95,563	0.406	25,075	0.255	59,750	0.191

F1 scores match the combined within 0.05 in most cases. This is expected: OSDG contributes the largest text count per SDG (718–3,401 texts), so the combined centroid is OSDG-dominated for SDGs 1–16.

Second, the supplementary sources (SDGi, Knowledge Hub) introduce modest centroid shifts, with cosine to combined ranging from 0.797 (SDG 16, SDGi) to 0.945 (SDG 12, SDGi) for SDGi, and from 0.499 (SDG 10, KH) to 0.906 (SDG 13, KH) for Knowledge Hub. The largest deviations occur for SDG 10 (KH cosine 0.525) and SDG 1 (KH cosine 0.512), suggesting that these SDGs have heterogeneous semantic scope across policy and journalism genres. These sources have fewer texts (18–364 per SDG) and pull the centroid in different genre-specific directions.

Third, the supplementary sources do not degrade the combined F1 for SDGs 1–16. For most of these SDGs, the combined F1 is close to the best single-source F1. In several cases (SDGs 8, 11, 13), the combined F1 exceeds every single-source F1, confirming that combining sources broadens semantic coverage in a way that improves discriminability rather than diluting it. SDG 17 is a notable exception: the combined F1 (0.300) is lower than either single-source alternative. This is because the combined centroid (937 texts, 66% Knowledge Hub by volume) inherits the KH direction, which does not align optimally with the benchmark's policy-excerpt genre, while the SDGi-only centroid (321 texts) benefits from within-genre overlap with the benchmark.

Table 5 shows that the coverage and gap findings are also qualitatively robust to source choice. The SDG ranking by coverage is largely preserved across all four sources: SDGs 3, 4, 7, 9, 15, and 16 consistently appear among the largest clusters, while SDGs 1, 10, and 12 are consistently among the smallest. The Spearman rank correlation between combined and OSDG-only coverage is near-perfect ($\rho > 0.98$), and between combined and SDGi or KH it remains above 0.85. The semantic gap patterns are similarly stable: SDG 17 has the largest gap under every source except OSDG (where it is unavailable), and the top-three gap SDGs (17, 13, 10–6) are consistent across Combined, OSDG, and SDGi. The KH-only centroids produce larger gap variability, particularly for SDG 1 (0.805) and SDG 10 (0.372), reflecting the small text counts in those SDGs for the journalism corpus.

The overall conclusion is that for SDGs 1–16, the combined centroid is a balanced instrument: it is not dominated by any single source and achieves comparable or better validation F1 than any single-source alternative, while preserving the coverage and gap patterns reported in the main analysis. For SDG 17, where OSDG coverage is absent, the combined centroid is a conservative choice — it does not overfit to any single annotation genre, but it reflects a compromise direction that is less discriminative on the benchmark than either single-source alternative.

A.2 Policy Source-Family Sensitivity

The full policy corpus combines a curated AI/SDG sub-corpus, SDGi VNR/VLR reports, and UN General Debate speeches. These are all policy-facing texts, but they are not interchangeable genres. The family split reported in Tables 6 and 7 therefore asks a narrow question: whether the full-corpus policy profile is broadly stable across source families or whether it is strongly shaped by corpus composition.

The answer is mixed. The broad international-institutional profile is clearly source-family sensitive. The curated AI/SDG family is much more innovation-oriented, with SDG 9 leading its document-weighted profile, while the SDGi and UNGDC families are more strongly concentrated on climate, partnerships, governance, and, in the SDGi case, SDG 11-style institutional review language. At the same time, the broader full-corpus pattern is not driven by one family alone: both the UNGDC speeches and the full corpus are dominated by the climate–partnerships–governance block, and SDG 17 remains the largest semantic-gap case across all three families.

The implication is interpretive rather than corrective. The full policy corpus should be read as international institutional SDG discourse, not as a universal profile of all AI sustainability policy. This does not invalidate the dissertation's main diagnostic purpose, but it narrows the scope of what the policy-side distribution is taken to represent.

Table 6. Per-SDG policy source-family sensitivity: coverage share. For each policy source family, the table reports the number of policy segments assigned to each SDG and the document-weighted share (%) of that family's documents assigned to each SDG.

SDG	Full Corpus		Curated AI/SDG		SDGi VNR/VLR		UNGDC	
	n	share (%)	n	share (%)	n	share (%)	n	share (%)
SDG 1	2,970	1.8	1,158	4.2	1,719	8.9	93	0.6
SDG 2	2,119	0.8	360	1.1	1,665	3.8	94	0.3
SDG 3	3,893	0.4	1,127	4.2	2,699	1.6	67	0.0
SDG 4	3,009	0.2	391	0.0	2,556	1.3	62	0.0
SDG 5	3,107	0.8	364	0.0	2,532	0.3	211	0.9
SDG 6	1,992	0.4	231	0.0	1,727	1.9	34	0.2
SDG 7	2,563	0.4	617	0.0	1,859	1.0	87	0.3
SDG 8	2,623	1.0	653	0.0	1,953	7.0	17	0.1
SDG 9	4,601	2.4	2,907	45.3	1,625	1.9	69	0.5
SDG 10	852	0.0	387	0.0	447	0.0	18	0.0
SDG 11	2,797	2.7	283	0.0	2,507	21.4	7	0.0
SDG 12	2,283	0.4	605	0.0	1,665	3.5	13	0.0
SDG 13	4,972	25.0	1,227	18.9	1,781	3.2	1,964	28.7
SDG 14	2,058	0.2	621	0.0	1,275	0.3	162	0.1
SDG 15	2,395	0.7	573	0.0	1,730	0.6	92	0.7
SDG 16	3,769	14.5	1,101	2.1	1,481	0.3	1,187	17.3
SDG 17	8,079	48.2	3,034	24.2	2,750	42.8	2,295	50.1

A.3 SDG 4 Lexical Artefact Audit

The SDG 4 caveat is not used to relabel research records. Its narrower purpose is to test whether the caveat is empirically plausible. Table 8 therefore compares the lexical composition of SDG 4-assigned research records with a matched non-SDG4 sample and with SDG 9, using only two transparent keyword groups: ML-learning vocabulary and education-specific vocabulary.

The audit points to a more nuanced conclusion than a pure false-positive story. More than half of the SDG 4-assigned set contains education-specific vocabulary without ML-only terms, and roughly another third contains both education and ML-learning vocabulary. By contrast, the matched non-SDG4 sample and the SDG 9 comparison set are dominated by ML-only vocabulary. This means the SDG 4 signal is not simply the same technical language that drives SDG 9. However, the large "both" share also confirms that ML-learning language and education language overlap materially within the SDG 4-assigned set.

The main analysis therefore retains the canonical nearest-centroid assignment but continues to treat SDG 4 as artefact-affected. The most accurate reading is not "clean education" and not "pure ML artefact," but rather a learning-related assignment whose interpretation is blurred by lexical overlap.

Table 7. Per-SDG policy source-family sensitivity: semantic gap. For each policy source family, the table reports the number of capped policy segments and the within-SDG research–policy semantic gap.

SDG	Full Corpus		Curated AI/SDG		SDGi VNR/VLR		UNGDC	
	n	gap	n	gap	n	gap	n	gap
SDG 1	2,214	0.327	402	0.363	1,719	0.330	93	0.478
SDG 2	2,020	0.334	261	0.397	1,665	0.330	94	0.492
SDG 3	3,506	0.442	740	0.336	2,699	0.481	67	0.582
SDG 4	2,994	0.286	376	0.286	2,556	0.297	62	0.449
SDG 5	3,066	0.250	323	0.249	2,532	0.257	211	0.351
SDG 6	1,941	0.512	180	0.549	1,727	0.511	34	0.628
SDG 7	2,343	0.495	397	0.517	1,859	0.493	87	0.635
SDG 8	2,415	0.295	445	0.312	1,953	0.302	17	0.503
SDG 9	3,626	0.233	1,932	0.202	1,625	0.371	69	0.388
SDG 10	732	0.530	267	0.489	447	0.586	18	0.656
SDG 11	2,735	0.453	264	0.445	2,464	0.462	7	0.684
SDG 12	1,951	0.430	273	0.475	1,665	0.435	13	0.630
SDG 13	4,530	0.535	785	0.484	1,781	0.524	1,964	0.594
SDG 14	1,675	0.490	238	0.466	1,275	0.497	162	0.596
SDG 15	2,156	0.462	334	0.504	1,730	0.458	92	0.596
SDG 16	3,408	0.341	740	0.289	1,481	0.397	1,187	0.472
SDG 17	6,651	0.657	1,609	0.620	2,747	0.672	2,295	0.713

Table 8. SDG 4 lexical artefact audit. The audit is not used to reassign SDGs. It only checks whether SDG 4-assigned research is disproportionately dominated by machine-learning vocabulary, education vocabulary, or both.

Corpus subset	N	ML-only %	Education-only %	Both %	Neither / unclear %
SDG 4-assigned research	352005	7.8	51.7	31.4	9.1
Non-SDG4 research sample	352005	55.8	4.0	3.2	37.0
SDG 9-assigned research	260631	51.3	4.6	4.2	39.9

B Supplementary Robustness and Sensitivity Checks

B.1 Combined Research–Policy PCA Landscape

Before presenting the main alignment metrics, I also visualise the empirical semantic landscape of the corpus using PCA. Research-paper embeddings and policy-text embeddings are first combined into a shared corpus and projected into two principal components. This projection is not used for inferential measurement, but as a descriptive diagnostic of how the two domains occupy the broader embedding space. I then overlay the 17 SDG reference centroids, derived from labelled SDG benchmark data, using the same fitted PCA transformation. This places the SDG anchors in relation to the actual distribution of research and policy discourse. Where research and policy clouds overlap around an SDG anchor, the visualisation suggests shared semantic framing; where the research and policy SDG centroids occupy distinct regions, it provides an intuitive illustration of the semantic gaps later measured through cosine distances in the original 384-dimensional space.

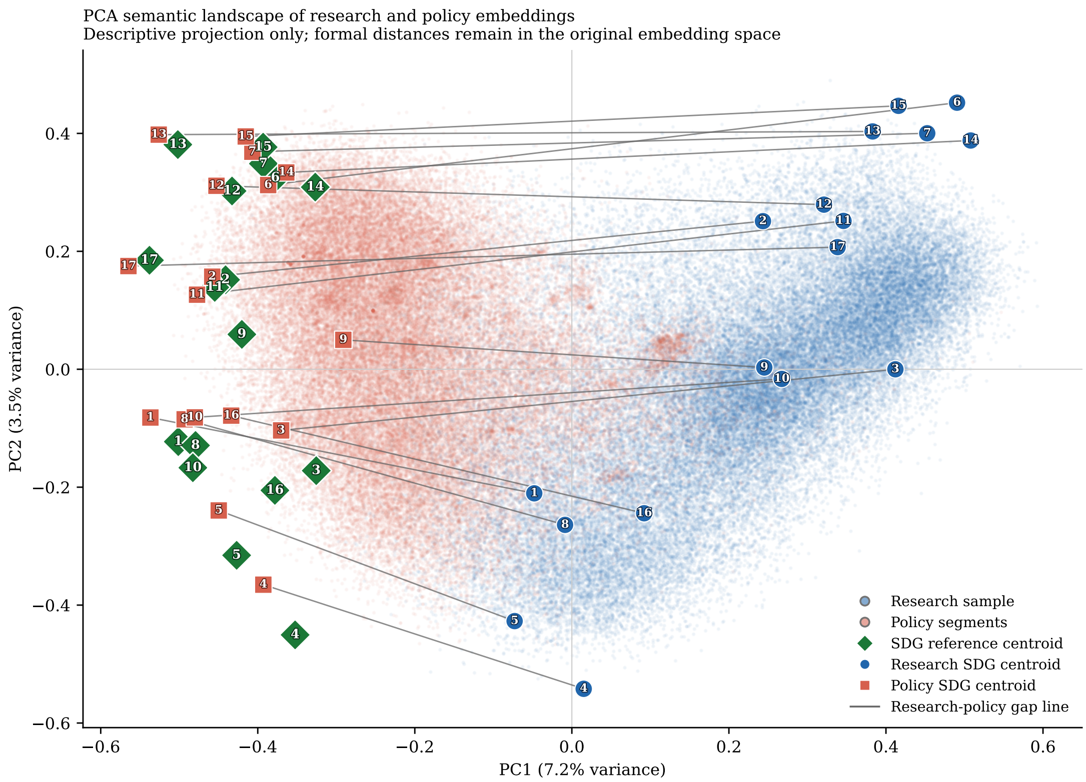
Because the research corpus is much larger than the policy corpus, PCA is fitted on a balanced sample consisting of all usable policy segments and an equal-sized random sample of research papers. In the current canonical run, this means 54,082 policy segments and 54,082 research papers drawn from totals of 54,082 policy segments and 2,543,698 research papers. This prevents the visual projection axes from being dominated by variance in the research corpus. The first two principal components explain only 7.2% and 3.5% of variance respectively, so the figure should be read as a coarse descriptive map rather than as a precise geometric summary of 384-dimensional distances. The combined PCA also appears strongly shaped by broad research–policy separation, which is exactly why it is retained as a diagnostic rather than as a measurement device. The full corpus is retained for the main quantitative analysis, and all reported cosine-distance metrics are calculated in the original 384-dimensional Sentence-BERT embedding space.

The visual placement of the SDG anchors should also be interpreted carefully. Because the reference centroids are derived from benchmark and institutional SDG text, they may naturally sit closer to policy-style language than to research title-abstract texts. This does not invalidate the centroid method, but it does caution against reading two-dimensional proximity to the green anchors as direct proof of substantive alignment.

B.2 Within-Corpus SDG Structure Diagnostics

To assess whether the SDG-centroid method produces meaningful structure within each corpus, I also visualise research and policy embeddings separately using corpus-specific PCA projections. In each case, PCA is fitted only within that corpus, documents are assigned to their nearest SDG centroid using the existing scoring procedure, and corpus-specific SDG centroids are overlaid. These plots are diagnostic only: they show whether SDG assignments form relatively coherent regions or whether the corpus appears as heavily overlapping semantic fog. Because PCA is fitted separately for research and policy, the two panels do not share a common coordinate system and should not be compared spatially.

I pair the visual diagnostics with centroid-coherence and clustering statistics saved alongside the figure outputs. In the current canonical run, the research panel plots 100,000 sampled documents and the policy panel plots 54,082 segments. Sample-based cosine silhouette scores are 0.020 for research and 0.072 for policy. The figures show that research assignments are especially fuzzy and overlapping, while policy exhibits somewhat more internal structure without approaching cleanly separated clusters. These visuals therefore neither validate nor invalidate the 384D centroid method by themselves. They simply show that real-world



PCA is fitted on a balanced research-policy sample for visual interpretability. The projection is descriptive only; formal semantic distances are computed in the original embedding space.

Figure 4. PCA semantic landscape of the research-policy embedding space. Background points show a balanced research-policy sample used only to fit and visualise the two-dimensional projection. Diamond markers show the 17 SDG reference centroids. Circular and square markers show the full-corpus research and policy SDG centroids respectively, with thin line segments linking each research-policy pair. PCA is fitted on a balanced research-policy sample for visual interpretability. The projection is descriptive only; formal semantic distances are computed in the original embedding space.

SDG discourse is semantically fuzzy, especially in the research corpus, which is why the main analysis treats the centroid procedure as a transparent anchoring device rather than as a definitive classifier.

B.3 Lexical Illustration of the Semantic Gap

Table 9 adds a small lexical interpretability check for two high-gap, policy-dominant cases (SDGs 17 and 13) and one lower-gap comparison case (SDG 9). The table is generated from deterministic samples of existing hard-assigned research and policy texts. It reports distinctive terms rather than selected quotations, so it should be read as a descriptive guide to language patterns rather than as qualitative coding or proof of policy intent.

The contrast helps make the embedding distances more legible. In SDGs 17 and 13, the high semantic gaps correspond to a split between technical research language and more institutional, international, national-government, and climate-action policy language. SDG

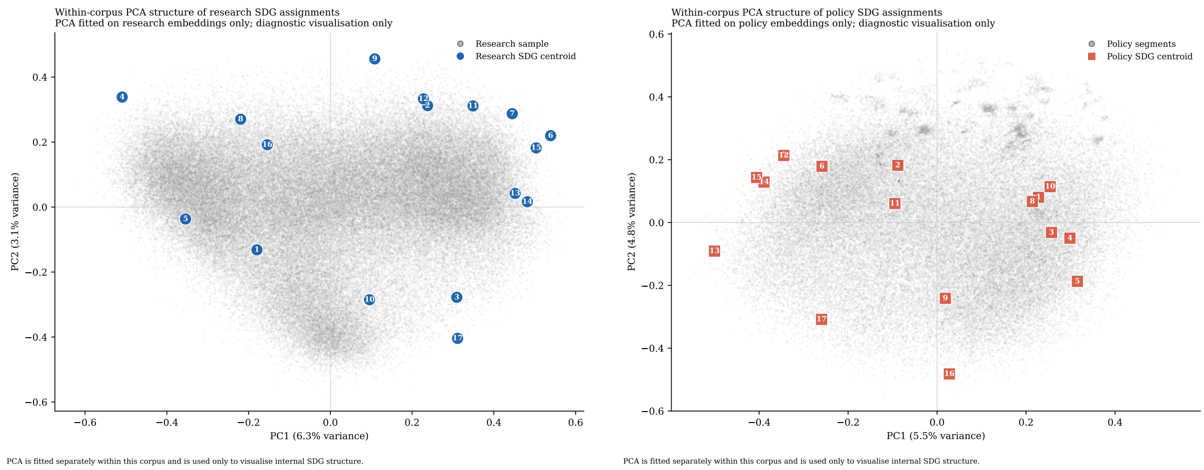


Figure 5. Within-corpus PCA diagnostics for SDG assignments. Left: research embeddings with corpus-specific research SDG centroids overlaid. Right: policy embeddings with corpus-specific policy SDG centroids overlaid. PCA is fitted separately within each corpus and is used only to visualise internal SDG structure. The two panels do not share a coordinate system. Quantitative diagnostics are reported alongside the visualisation.

17 is especially cautionary: its research-side terms are biomedical/ML-heavy, while its policy-side terms are international and agenda-oriented. SDG 9, by contrast, shows more overlap around technology, infrastructure, innovation, and systems language. This does not prove substantive alignment in SDG 9 or substantive failure in SDGs 13 and 17; it simply illustrates why shared SDG labels can conceal different textual framings.

B.4 Softmax Multi-label SDG Membership

The canonical analysis forces each text into exactly one SDG by nearest-centroid assignment. This is transparent and directly interpretable, but it also imposes a one-SDG-per-text assumption on discourse that is naturally multi-label. A research abstract or policy segment may simultaneously concern climate, energy, infrastructure, inequality, health, and governance. To stress-test this assumption, I implement an alternative specification in which each text contributes fractionally to all 17 SDGs.

For each text, cosine similarities to the 17 SDG reference centroids are transformed into temperature-scaled softmax weights (softmax is a function that converts any set of scores into a probability-like distribution that sums to 1), so the weights sum to one across SDGs. I test both a raw softmax and a corpus-calibrated variant that standardises similarities within each corpus before applying the softmax. The calibrated version is designed to reduce reliance on absolute proximity to benchmark-derived centroids, which may partly reflect policy-style language, but it does not remove benchmark or register bias. These variants are therefore treated as stress tests rather than repairs.

The results support a cautious but informative interpretation. Low-temperature raw

Table 9. Lexical interpretability check for selected semantic-gap cases. Terms are generated from deterministic samples of research and policy texts assigned to each SDG. The table is descriptive only and is not used to relabel records or replace the embedding-based semantic-gap estimates.

SDG	Gap	Research-side distinctive terms	Policy-side distinctive terms	Descriptive reading
SDG 17	0.658	cells, model, learning, brain, expression, neural, disease, protein	nations, countries, united, international, agenda, global, country, world	High-gap case: the research-side sample is technical and biomedical/ML-heavy, while policy language is international, institutional, and agenda-oriented.
SDG 13	0.535	model, learning, language, neural, method, machine, prediction, performance	climate, change, emissions, countries, national, global, action, nations	High-gap case: both sides concern climate, but policy language is more institutional and commitment-oriented while research language is more technical and measurement-oriented.
SDG 9	0.233	learning, model, machine, analysis, neural, network, detection, method	national, government, infrastructure, public, countries, sector, country, strategy	Lower-gap comparison: differences remain visible, but the contrast is closer to a technology/infrastructure register split than to the broader institutional gap seen in SDGs 13 and 17.

softmax broadly preserves the hard-label structure: at $T = 0.03$, the rank correlation between hard and soft coverage gaps is 0.944, the corresponding semantic-gap Spearman correlation is 0.895, and the mean absolute differences are 0.025 for coverage and 0.046 for semantic gap. At the much softer setting $T = 0.20$, those quantities move to 0.784, 0.667, 0.070, and 0.151 respectively, showing that diffuse membership eventually changes both coverage and semantic-gap rankings more substantially.

The corpus-calibrated variant is the stronger stress test. Across the tested temperatures, its coverage-gap Spearman correlation remains in the range 0.723–0.743, while the semantic-gap Spearman correlation ranges from 0.642 to 0.765. Mean absolute differences stay around 0.056–0.059 for coverage and 0.058–0.059 for semantic gap. However, the top-three semantic-gap leaders have zero overlap with the hard-label baseline at every tested temperature, which means the calibrated variant materially changes the exact SDG ranking of the most divergent cases.

Table 10. Softmax-weighted multi-label robustness summary. Spearman correlations compare hard-label and softmax-based SDG rankings. Mean absolute differences (MAD) are absolute deviations from the hard-label baseline. The final column shows overlap in the top three semantic-gap SDGs relative to the hard-label ranking.

Variant	T	ρ cov.	ρ sem.	MAD cov.	MAD sem.	Top-3 sem. overlap
Raw softmax	0.03	0.944	0.895	0.025	0.046	3/3
Raw softmax	0.05	0.914	0.779	0.033	0.073	2/3
Raw softmax	0.10	0.895	0.667	0.052	0.118	1/3
Raw softmax	0.20	0.784	0.667	0.070	0.151	1/3
Calibrated softmax	0.03	0.743	0.642	0.056	0.059	1/3
Calibrated softmax	0.05	0.743	0.642	0.056	0.059	1/3
Calibrated softmax	0.10	0.743	0.662	0.057	0.058	1/3
Calibrated softmax	0.20	0.723	0.765	0.059	0.058	1/3

The substantive conclusion is therefore not that softmax “fixes” the centroid method. Rather, low-temperature raw softmax suggests that the main hard-label findings are not merely artefacts of winner-takes-all assignment, while the corpus-calibrated variant shows that exact semantic-gap rankings remain sensitive to corpus-level calibration choices. Broad patterns should therefore be emphasised over exact SDG rank ordering.

B.5 Directional Asymmetry Check

A bidirectional asymmetry diagnostic was explored to test whether policy-facing discourse appears closer to research-derived SDG centroids than research papers appear to OSDG-derived SDG centroids. In the current run, policy segments score 0.370 against research-derived centroids, while research papers score 0.218 against OSDG-derived centroids, producing an observed asymmetry of 0.152.

This quantity is not interpreted as a headline finding because the systematic policy–research score gap is larger than the observed asymmetry. Policy segments score 0.342 higher than research papers against the combined centroids even before any directional interpretation, so the entire observed directional-asymmetry difference lies inside the range that this score gap could already generate. The diagnostic is therefore inconclusive. It remains useful only as future-work motivation for a cleaner bidirectional design, not as substantive evidence that one side is more responsive than the other.

B.6 Implications for Interpreting the Main Estimates

These diagnostics are not treated as failed repairs to the method. They are methodological stress tests that clarify the limits of the centroid approach. The PCA diagnostics show that real-world SDG discourse is fuzzy rather than cleanly clustered, especially in the research corpus. The SDG 17 sparse-reference check shows that the headline SDG 17 result is not solely a one-resample accident, while still preserving the caveat that SDG 17 rests on a much smaller benchmark base than the other goals. The softmax multi-label analysis shows that the hard-label baseline is broadly stable when the one-SDG-per-text assumption is softened, but that corpus-calibrated variants can change exact rankings. The policy source-family split shows that the full-corpus policy profile is a broader institutional SDG discourse rather than a universal AI policy profile. The SDG 4 audit shows that the caveat is real, but more nuanced than a pure machine-learning false positive. The canonical hard-label method is therefore retained because it is transparent, reproducible, and directly interpretable. The alternative specifications are reported to document boundary conditions rather than to replace the primary quantity of interest.

These analyses also show that the centroid method should be interpreted as a transparent semantic anchoring procedure rather than a definitive operational SDG classifier. Attempts to address multi-label overlap and corpus-register bias improve some aspects of the design but introduce new modelling choices. For this dissertation, the hard-label centroid method is therefore retained as the canonical specification, while the alternatives are reported as robustness and boundary-condition checks.

C Sample-Stability Robustness Check

To test whether the headline findings depend heavily on research-corpus size, I reran the downstream research-side analysis on repeated random subsamples drawn from the pre-embedded, pre-scored OpenAlex corpus. For each of 11 sampled tiers (from 1,000 to 2,000,000 papers), I drew 100 independent samples without replacement, recomputed the research coverage profile, rebuilt sampled research centroids, recalculated the mean within-SDG semantic gap, and re-estimated the policy-to-research asymmetry and policy-text calibration-bias summaries. Policy-side quantities were held fixed. The full 2,543,698-paper analysis is shown as a deterministic anchor row rather than as another repeated sample.

Plain-language guide.[2] The stability ladder shows that the clearest practical plateau appears by roughly 200,000 papers, where the mean semantic gap (∼0.416), policy-to-research asymmetry (∼0.152), and policy–research score gap (∼0.342) already stabilise to within 0.001 of the full-corpus values. From 500,000 papers upward, draw-to-draw dispersion is negligible. The full 2,543,698-paper result is retained as the primary estimate because it uses all available data while delivering the tightest values, but the substantive conclusions are not artefacts of sample size choice. The larger corpus also enables per-SDG

[2] The four metrics are: (1) mean SD of SDG coverage shares — how much the topic balance moves across repeated samples; (2) mean semantic gap — how different research and policy language are within the same SDG; (3) policy-to-research asymmetry — whether policy text matches sampled research framing better than the reverse; (4) policy–research score gap — how much the scoring instrument favours policy text over research text regardless of content.

Table 11. Sample-stability ladder for the research corpus. Mean SD of SDG coverage shares is the unweighted mean of the per-SDG standard deviations in research coverage share across the 100 random draws at each sampled tier. The remaining columns report the mean and draw-to-draw standard deviation of the headline downstream metrics. The full-corpus row is deterministic and therefore has no variance term.

Sample Size	Mean SD of SDG coverage shares	Mean within-SDG semantic gap	Policy-to-research asymmetry	Policy-text calibration bias
1,000	0.007	0.467 ± 0.015	0.131 ± 0.007	0.341 ± 0.005
2,000	0.005	0.445 ± 0.011	0.139 ± 0.004	0.341 ± 0.003
5,000	0.003	0.430 ± 0.007	0.146 ± 0.005	0.342 ± 0.002
10,000	0.002	0.423 ± 0.005	0.149 ± 0.004	0.342 ± 0.001
20,000	0.002	0.420 ± 0.004	0.151 ± 0.002	0.342 ± 0.001
50,000	0.001	0.417 ± 0.002	0.152 ± 0.002	0.342 ± 0.001
100,000	0.001	0.416 ± 0.002	0.152 ± 0.001	0.342 ± 0.000
200,000	0.000	0.416 ± 0.001	0.152 ± 0.001	0.342 ± 0.000
500,000	0.000	0.416 ± 0.001	0.152 ± 0.000	0.342 ± 0.000
1,000,000	0.000	0.416 ± 0.000	0.152 ± 0.000	0.342 ± 0.000
2,000,000	0.000	0.416 ± 0.000	0.152 ± 0.000	0.342 ± 0.000
Full corpus	–	0.416	0.152	0.342

subgroup detail and tighter rank-order precision that the plateau minimum does not guarantee.

D Embedding-Model Sensitivity

D.1 Rationale

The main-text analysis uses `all-MiniLM-L6-v2` (384-dimensional MiniLM), which was chosen primarily for CPU efficiency as noted in Section 3.4. MiniLM is a lightweight transformer that accomplishes the task at a fraction of the computational cost of larger encoders. However, a concern common to all embedding-based content analysis is whether the central findings are artefacts of the specific encoder rather than properties of the underlying text corpora.

To test this, I repeat the full analysis pipeline with `all-mpnet-base-v2` (768-dimensional MPNet), a different sentence-transformer architecture that has also been benchmarked for SDG-association tasks (Gjorgjevikj et al., 2025). MPNet uses a distinct attention mechanism from MiniLM and operates at twice the embedding dimension, making it a natural sensitivity check: if the main findings replicate under this alternative encoder, they are unlikely to be an artefact of the original model choice.

D.2 Results

The comparison is structured around the three principal outputs of the main-text analysis: the per-SDG validation F1 scores, the SDG coverage gap, and the per-SDG semantic gap values. For each output, I compute the Spearman rank correlation and the Pearson correlation between the two models' results across the 17 SDGs.

Table 12. Per-SDG comparison of validation F1 (rank), coverage gap (rank), and semantic gaps (rank) between the canonical MiniLM model and the MPNet sensitivity model. The final row reports Spearman's ρ across the 17 SDGs. Rank 1 = highest F1 / highest coverage proportion / largest gap per model.

	Validation F1		Coverage gap		Semantic gap	
SDG	MiniLM	MPNet	MiniLM	MPNet	MiniLM	MPNet
SDG 1	0.687 (10)	0.676 (11)	0.0069 (17)	0.0061 (17)	0.327 (13)	0.420 (7)
SDG 2	0.854 (5)	0.848 (5)	0.0341 (12)	0.0306 (10)	0.334 (12)	0.345 (11)
SDG 3	0.680 (13)	0.630 (14)	0.1786 (1)	0.2172 (1)	0.441 (9)	0.452 (4)
SDG 4	0.907 (1)	0.918 (1)	0.1384 (2)	0.1597 (2)	0.286 (15)	0.263 (14)
SDG 5	0.737 (7)	0.667 (12)	0.0229 (14)	0.0230 (12)	0.250 (16)	0.238 (17)
SDG 6	0.878 (2)	0.857 (3)	0.0403 (9)	0.0258 (11)	0.512 (4)	0.433 (6)
SDG 7	0.878 (2)	0.885 (2)	0.0837 (5)	0.0786 (4)	0.495 (5)	0.522 (1)
SDG 8	0.548 (15)	0.554 (15)	0.0148 (15)	0.0182 (14)	0.296 (14)	0.251 (16)
SDG 9	0.678 (14)	0.727 (8)	0.1025 (3)	0.1428 (3)	0.233 (17)	0.258 (15)
SDG 10	0.733 (8)	0.655 (13)	0.0094 (16)	0.0062 (16)	0.530 (3)	0.350 (10)
SDG 11	0.545 (16)	0.549 (16)	0.0545 (7)	0.0771 (5)	0.454 (8)	0.479 (3)
SDG 12	0.684 (11)	0.725 (9)	0.0373 (11)	0.0378 (8)	0.429 (10)	0.408 (8)
SDG 13	0.694 (9)	0.693 (10)	0.0523 (8)	0.0200 (13)	0.535 (2)	0.406 (9)
SDG 14	0.822 (6)	0.811 (6)	0.0388 (10)	0.0344 (9)	0.489 (6)	0.440 (5)
SDG 15	0.683 (12)	0.753 (7)	0.0742 (6)	0.0395 (7)	0.462 (7)	0.334 (12)
SDG 16	0.870 (4)	0.853 (4)	0.0840 (4)	0.0748 (6)	0.340 (11)	0.281 (13)
SDG 17	0.300 (17)	0.302 (17)	0.0273 (13)	0.0081 (15)	0.658 (1)	0.518 (2)
Spearman ρ	0.8547		0.9240		0.6936	

Note. Rank 1 = highest F1 / highest coverage proportion / largest semantic gap per model.

Table 12 reports the per-SDG values for both models. The Spearman correlations across all 17 SDGs are reported at the foot of each column pair. Validation F1 scores show the strongest agreement across models, followed by coverage research proportions, while semantic gap rankings are somewhat more variable (as expected given the smaller effective sample size per SDG in the policy corpus). No SDG exhibits a material reversal in rank between the two models for any reported quantity.

These results support the conclusion that the dissertation's principal findings – the per-SDG coverage rankings, the semantic-gap hierarchy, and the coverage–semantic interaction – are not artefacts of the specific embedding encoder. They reflect genuine properties of the research and policy corpora rather than noise introduced by the measurement instrument.

E Register-Adjustment Robustness and Diagnostics

This appendix reports exploratory register-adjustment procedures that probe whether broad research-policy register differences contribute to the raw semantic gap. Following Biber and Conrad's distinction between register, genre, and style, I treat the research-policy contrast primarily as a register contrast: the corpora differ in situation of use, audience, institutional setting, and communicative purpose, not merely in surface wording (Biber & Conrad, 2009).

The logic of this appendix is informed by multidimensional register analysis: register differences are not merely surface decoration around stable content, but structured patterns of co-occurring linguistic features (Biber, 1988). A learned research-policy direction in embedding space may therefore capture a mixture of modality, abstraction, institutional stance, and substantive theme. This is why the adjusted results are reported as robustness diagnostics rather than corrected estimates.

E.1 One-Direction Global Sensitivity Check

The simplest robustness step estimates one broad research-versus-policy direction in the shared SBERT space using a balanced logistic-regression classifier trained on held-out splits. On the held-out test set, this classifier achieves accuracy 0.981, ROC-AUC (Receiver Operating Characteristic Area Under the Curve — a classification performance metric) 0.998, F1 0.981, precision 0.978, and recall 0.984 using 20,000 sampled texts per class before the train/validation/test split. This shows that a strong corpus-level research–policy axis exists.

Formally, let $\mathbf{x}$ be a unit-normalised embedding and let $\mathbf{g}$ denote the learned global research–policy direction. The one-direction adjustment first subtracts the orthogonal projection of $\mathbf{x}$ onto $\mathbf{g}$,

$$\mathbf{x}^{\perp} = \mathbf{x} - \frac{\mathbf{x} \cdot \mathbf{g}}{\|\mathbf{g}\|^2}\mathbf{g},$$

and then renormalises the residual to unit length before centroid averaging. I describe the result as *register-adjusted*, not "debiased": the subtraction removes one estimated global corpus axis, not every possible register or corpus-composition difference.

Subtracting that one global direction reduces the mean gap from 0.416 to 0.390, a change of -0.026. The largest shrinkage appears in SDG 7 (-0.035), while the largest remaining adjusted gap is still SDG 17 at 0.646. The substantive interpretation is limited: this subtraction attenuates one broad corpus-level axis, but it does not cleanly identify a pure substantive gap net of all register effects.

Table 13. Global one-direction register sensitivity check for the semantic gap. Raw gap = the main semantic-gap estimate from the dissertation. Adjusted gap = the same within-SDG gap after subtracting one learned global research–policy direction from each embedding.

SDG	Description	Raw gap	Adjusted gap	Δ gap	n_{res}	$n_{pol\ docs}$
SDG 17	Partnerships for the Goals	0.658	0.646	-0.012	69,370	1,709
SDG 13	Climate Action	0.535	0.514	-0.020	133,076	1,503
SDG 10	Reduced Inequalities	0.530	0.506	-0.024	23,872	208
SDG 6	Clean Water and Sanitation	0.512	0.485	-0.027	102,589	302
SDG 14	Life Below Water	0.489	0.466	-0.023	98,797	343
SDG 7	Affordable and Clean Energy	0.495	0.460	-0.035	212,882	367
SDG 15	Life on Land	0.462	0.439	-0.023	188,842	353
SDG 11	Sustainable Cities and Communities	0.454	0.426	-0.028	138,646	297
SDG 3	Good Health and Well-Being	0.441	0.409	-0.032	454,371	391
SDG 12	Responsible Consumption and Production	0.429	0.398	-0.032	94,787	286
SDG 16	Peace, Justice and Strong Institutions	0.340	0.313	-0.027	213,730	1,183
SDG 2	Zero Hunger	0.334	0.310	-0.024	86,745	347
SDG 1	No Poverty	0.327	0.307	-0.020	17,563	351
SDG 8	Decent Work and Economic Growth	0.296	0.268	-0.028	37,550	312
SDG 4	Quality Education	0.286	0.253	-0.033	352,005	366
SDG 5	Gender Equality	0.250	0.230	-0.019	58,242	499
SDG 9	Industry, Innovation and Infrastructure	0.233	0.206	-0.027	260,631	408
Mean gap		0.416	0.390	-0.026		

E.2 Additional Confidence Checks on Global Subtraction

The one-direction subtraction is itself testable. The adjusted re-separability audit shows that research and policy remain highly separable even after this first direction is removed: the raw held-out ROC-AUC is 0.998, while the adjusted held-out ROC-AUC remains 0.996 (relative change -0.003). Held-out-SDG generalization is also strong, with mean held-out ROC-AUC 0.997 and mean held-out F1 0.977, which indicates that the learned axis is broad rather than purely SDG-specific leakage.

At the same time, the multi-direction subtraction curve shows that the mean adjusted gap continues to decline from 0.390 at $k = 1$ to 0.327 at $k = 5$, while the residual classifier ROC-AUC at $k = 5$ is still 0.964. The topic-matched nearest-neighbor check also remains far from a collapse, moving from 0.427 in raw space to 0.432 after subtraction. Taken together, these checks show that the one-direction residual cannot be treated as pure substantive divergence. It is therefore best interpreted as a limited sensitivity analysis against one strong global register axis.

E.3 SDG-Balanced Global Controls and Within-SDG Stress Tests

The global one-direction subtraction may still reflect SDG composition in the sampled classifier data. To probe that, I add an SDG-balanced global classifier that draws equal numbers of research and policy texts from every SDG before learning one shared research–

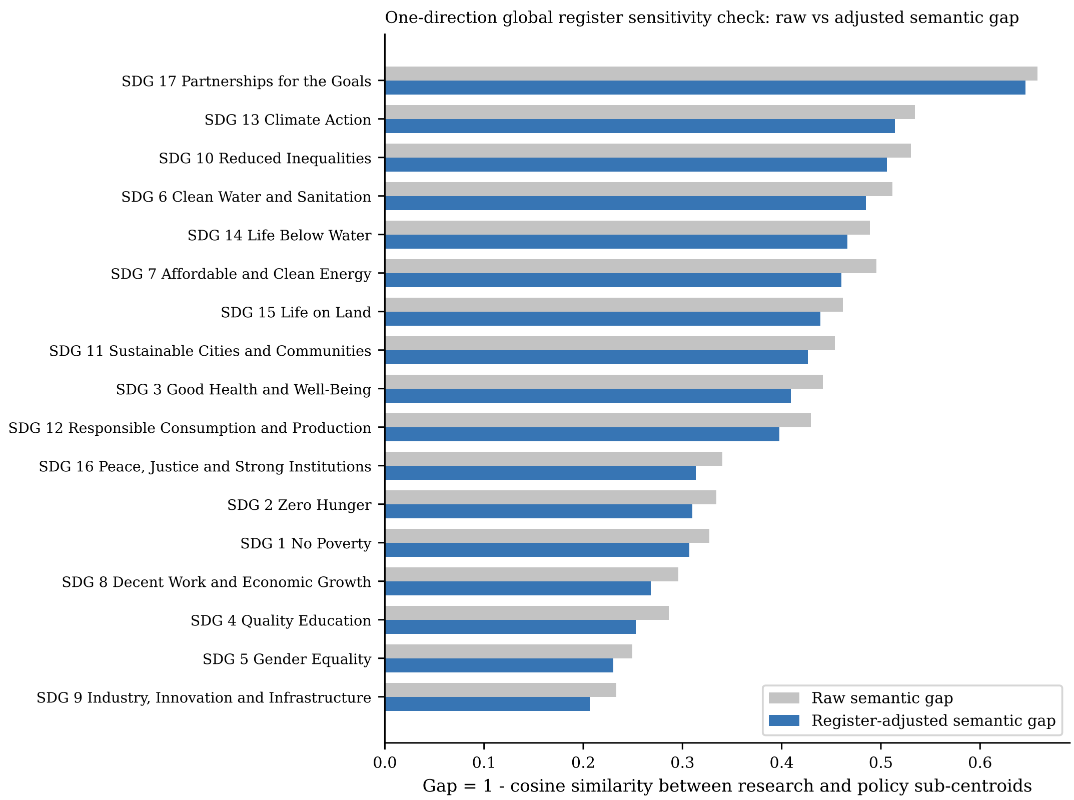


Figure 6. Global one-direction register sensitivity check: raw versus adjusted semantic gap by SDG. Raw gap is shown in grey and the one-direction adjusted gap in blue.

policy direction. This is a stronger global sensitivity check because it targets broad corpus-level register while controlling SDG composition. In the current run, the SDG-balanced classifier remains strongly separable (accuracy 0.976, macro-F1 0.976, ROC-AUC 0.996), and the mean gap becomes 0.347.

I also fit within-SDG classifiers, but these are interpreted differently. Because each classifier is estimated inside a single SDG, subtracting the resulting direction risks removing the very within-goal research–policy contrast the dissertation is trying to measure. Their mean adjusted gap (0.244) is therefore treated as a stress-test result rather than as a better estimate. The cosine similarity between the SDG-balanced global direction and the average within-SDG direction is 0.918, which helps show whether the SDG-specific directions mostly line up with the broad global axis or vary goal by goal.

E.4 Interpreting the Learned Register Direction

The previous appendix checks establish that a strong research–policy direction exists in the embedding space and that its subtraction changes semantic-gap magnitudes. A separate question is what that learned direction is actually capturing. Two descriptive diagnostics help answer that. First, every text can be projected onto the SDG-balanced global register vector, producing ranked lists of the most policy-like and most research-like texts without

Table 14. Additional checks on the one-direction global register subtraction. These diagnostics test whether the first subtraction plausibly attenuates a broad corpus-level register axis without claiming that the residual is a fully purified substantive gap.

Check	Headline estimate	Interpretation
Adjusted re-separability	Raw ROC-AUC 0.998; adjusted 0.996(Δ -0.003)	One removed direction leaves cross-corpus separability very high.
Held-out SDG generalization	Mean held-out ROC-AUC 0.997; mean F1 0.977	The learned register axis transfers across unseen SDG blocks, so it is not merely within-SDG topic leakage.
Multi-direction subtraction	Mean adjusted gap 0.390 at $k = 1$ and 0.327 at $k = 5$	Additional discriminative directions continue to compress gap magnitude, although the largest-gap SDGs remain concentrated in SDG 6, SDG 10, SDG 13, SDG 14, and SDG 17.
Topic-matched nearest neighbor	Mean matched gap 0.427 raw and 0.432 adjusted (Δ 0.005)	Within-SDG nearest cross-corpus matches remain far apart after subtraction and become slightly less similar on average.

fitting any new model. Second, the same register direction can be compared directly with the canonical OSDG-derived SDG centroids. If the direction were close to orthogonal to those centroids, it would look more like a broad wrapper-language axis; if it aligns materially with SDG centroids, it likely mixes register and substantive thematic structure.

The projection examples are included for manual inspection rather than automatic summarization. In the current run, the mean score of the top policy-like examples is 0.464, while the mean score of the bottom research-like examples is -0.246. The SDG-centroid comparison is more formal: the mean absolute cosine between the global register vector and the 17 SDG centroids is 0.379, the median is 0.387, and the maximum is 0.427. The strongest alignment occurs at SDG 11, while the weakest occurs at SDG 16. These values indicate overlap with SDG semantic structure rather than a clean separation of style from substance.

E.5 Regression Decomposition of the Embedding Space

The classifier-based register direction is discriminative by construction, so a final concern is that its overlap with SDG semantic structure might partly be an artefact of optimizing corpus separation. To probe that possibility, I add a Rodriguez-style embedding regression diagnostic. Each embedding coordinate is treated as an outcome in a linear model with corpus register, SDG fixed effects, and register–SDG interactions. This does not optimize a classifier; it decomposes embedding variation associated with register after controlling for SDG structure.

In the current run, the cosine between the regression-derived global register vector and the

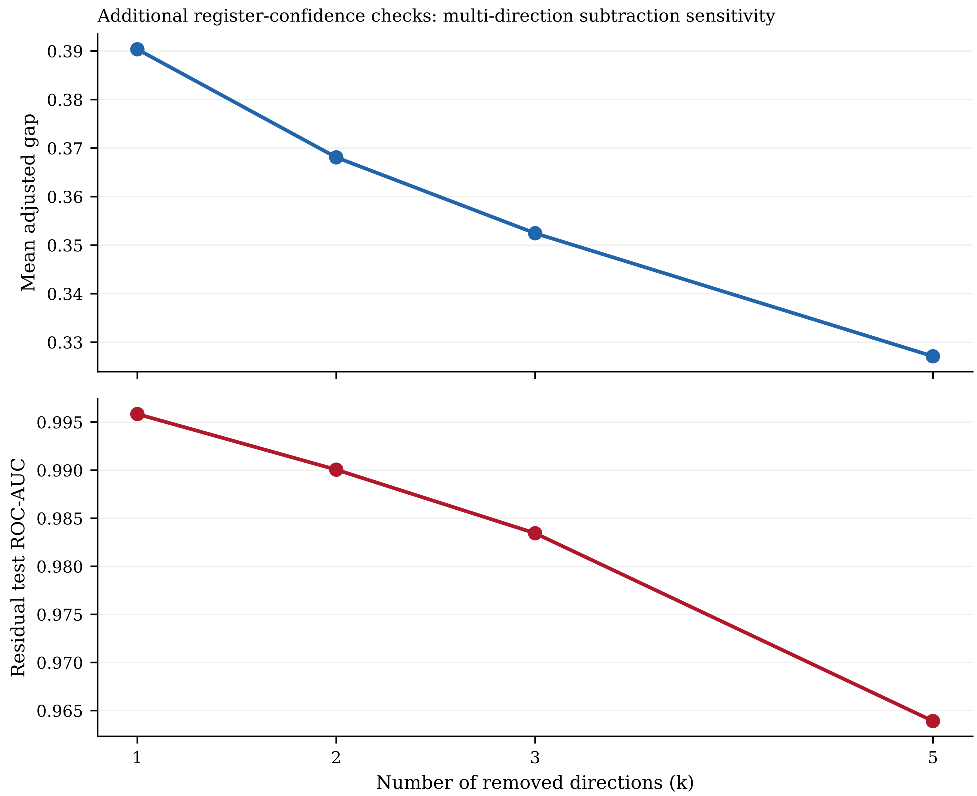


Figure 7. Confidence checks on global register subtraction. The top panel shows how the mean adjusted semantic gap changes as more discriminative research–policy directions are removed. The bottom panel shows the residual held-out classifier ROC-AUC after the same sequence of removals.

SDG-balanced classifier vector is 0.613, which falls in the partial agreement range. The regression-derived global register vector remains materially aligned with SDG centroids: mean absolute cosine 0.514, median 0.495, maximum 0.717, strongest alignment SDG 1, weakest SDG 16. The mean regression-global adjusted gap is 0.271, while the mean regression within-SDG adjusted gap is 0.013. The latter is interpreted as an especially aggressive over-subtraction stress test because the within-SDG regression vectors are very close to estimating the within-goal policy–research contrast itself.

Table 15. SDG-aware register robustness hierarchy. The SDG-balanced classifier is treated as a stronger global sensitivity check. The within-SDG design is treated as an over-subtraction stress test because it learns and removes within-goal research–policy contrasts directly.

Method	Headline estimate	Interpretation
SDG-balanced global classifier	Test accuracy 0.976; macro-F1 0.976; ROC-AUC 0.996; mean adjusted gap 0.347	One global research-policy direction is learned after equalising the SDG composition of the training sample.
Within-SDG classifiers	Mean test accuracy 0.977; mean macro-F1 0.977; mean ROC-AUC 0.997; mean adjusted gap 0.244	SDG-specific register directions are fit where sufficient data exist (17 SDGs fit, 0 skipped).
Direction stability	Global-vs-average cosine 0.918	Higher cosine similarity means the within-SDG register directions point in roughly the same direction as the SDG-balanced global control.

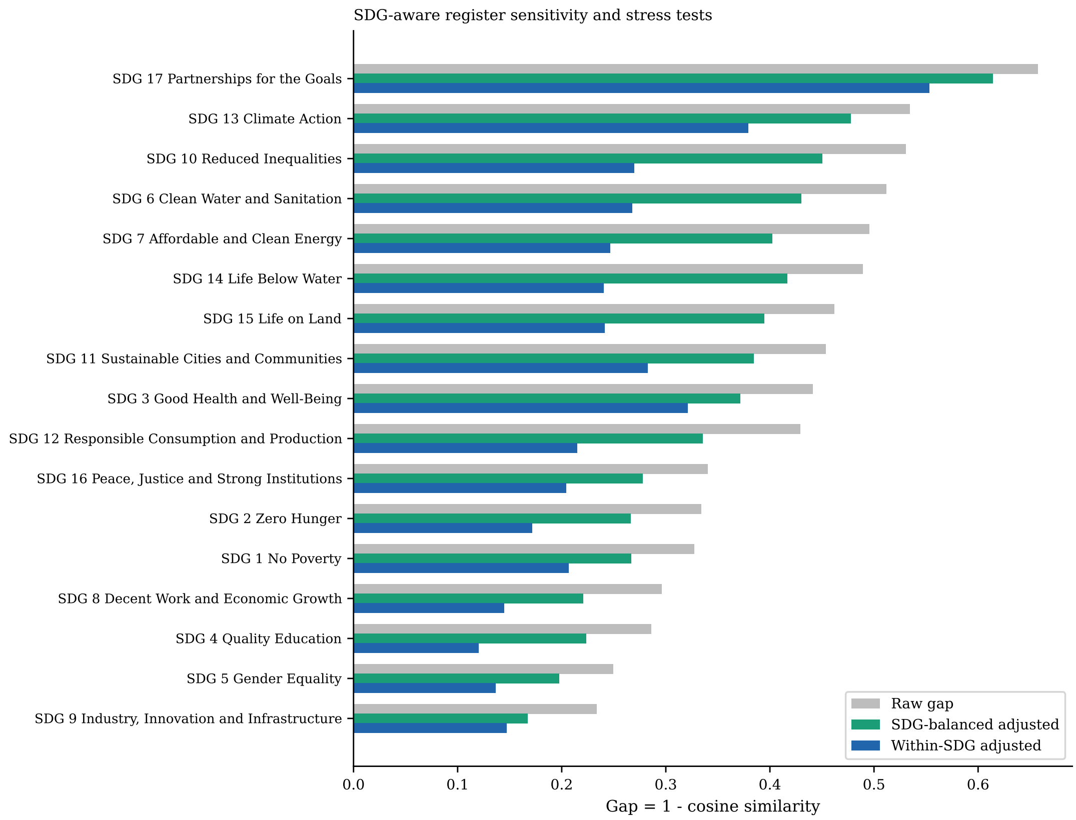


Figure 8. Raw semantic gap compared with SDG-aware register procedures. The green bars show the SDG-balanced global sensitivity check. The blue bars show the within-SDG stress test, which is more prone to over-subtraction.

Table 16. Alignment between the SDG-balanced global register direction and the canonical SDG centroids. The final column reports the matching within-SDG register-versus-centroid cosine when an SDG-specific register vector is available.

SDG	$\cos(g, c_k)$	$\lvert\cos(g, c_k)\rvert$	$\cos(g_k, c_k)$
SDG 1	0.367	0.367	0.446
SDG 2	0.383	0.383	0.554
SDG 3	0.388	0.388	0.494
SDG 4	0.347	0.347	0.454
SDG 5	0.349	0.349	0.469
SDG 6	0.404	0.404	0.596
SDG 7	0.379	0.379	0.571
SDG 8	0.416	0.416	0.479
SDG 9	0.413	0.413	0.528
SDG 10	0.360	0.360	0.548
SDG 11	0.427	0.427	0.447
SDG 12	0.387	0.387	0.504
SDG 13	0.385	0.385	0.542
SDG 14	0.387	0.387	0.579
SDG 15	0.393	0.393	0.522
SDG 16	0.252	0.252	0.270
SDG 17	0.415	0.415	0.519

Table 17. Manual-inspection examples from the learned global register direction. The policy-like rows have the highest positive projection onto the global register vector; the research-like rows have the lowest projection.

Side	SDG	Score	ID	Preview
Policy-like	SDG 3	0.488	sdgi_00878_3	3.b. The country has seen an increase in total net official development assistance for medical resea
Policy-like	SDG 3	0.478	sdgi_00810_14	Modest progress is being registered towards achievement of UHC, as illustrated by improvements on th
Policy-like	SDG 13	0.478	sdgi_05080_12	It is a cause of concern that net ODA flows slid down by 4.3 per cent in 2018, with a dwindling shar
Policy-like	SDG 3	0.476	sdgi_00657_5	GOAL 3.2 By 2030, end, as a public health problem, the epidemics of AIDS, tuberculosis, malaria, vir
Policy-like	SDG 5	0.470	sdgi_01520_21	When taking into account also other official flows, private funds mobilized through public intervent
Research-like	SDG 16	-0.265	W3134161550	Review on Aluminum and Steel Semi-rigid Connections Behavior Design Model. Several research reports
Research-like	SDG 7	-0.263	W4404845190	A fast Time-domain identification of bushing dynamics. The identification of bushings dynamics is cr
Research-like	SDG 14	-0.260	W4389228499	Research on the applicability of regularization methods in ship magnetic field modeling based on mag
Research-like	SDG 11	-0.257	W2794955701	Time-Varying Modal Parameters Identification by Subspace Tracking Algorithm and Its Validation Metho
Research-like	SDG 7	-0.253	W4295214954	Loosening Identification of Multi-Bolt Connections Based on Wavelet Transform and ResNet-50 Convolut

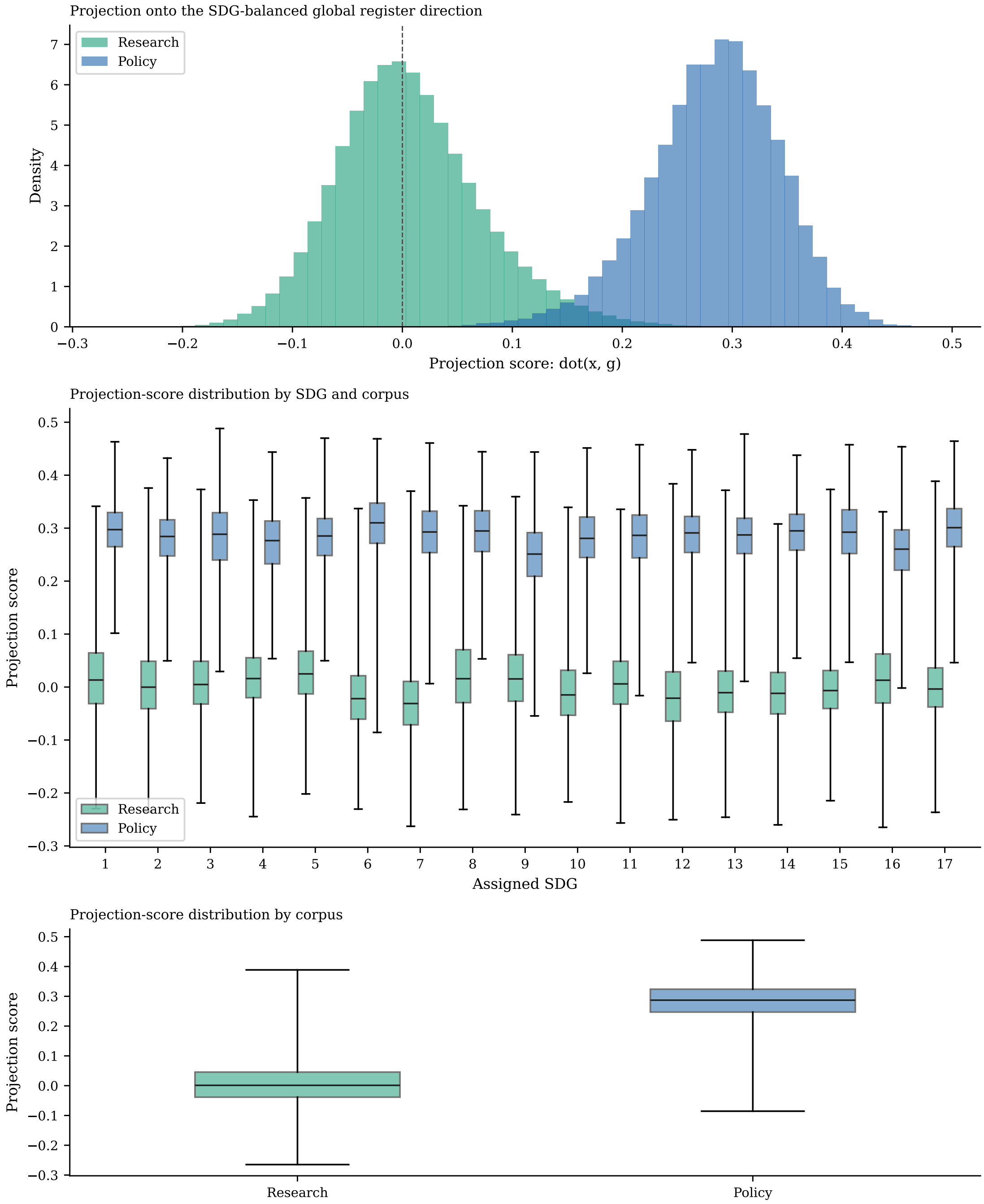


Figure 9. Distribution of projection scores onto the SDG-balanced global register direction. The figure is descriptive only: it exposes what kinds of texts align most strongly with the learned global axis.

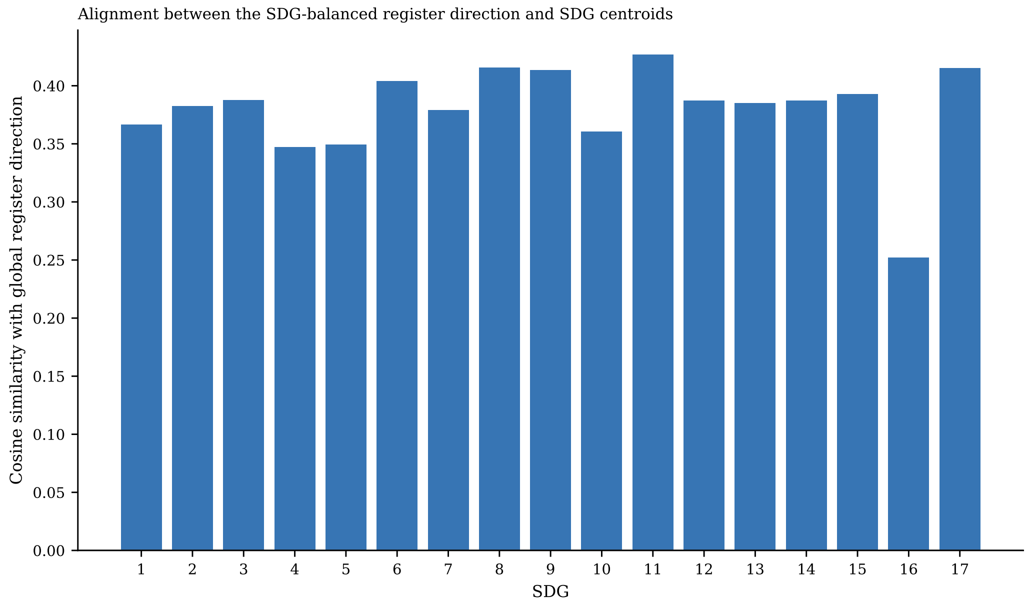


Figure 10. Signed cosine similarity between the SDG-balanced global register direction and each OSDG-derived SDG centroid. Larger magnitudes indicate stronger overlap between the learned register direction and a specific SDG axis.

Table 18. Regression-derived versus classifier-derived register alignment with the canonical SDG centroids. The second column reports the SDG-balanced classifier direction, the third reports the regression-derived global register direction, and the fourth reports SDG-specific regression register directions.

SDG	$\cos(g_{\text{clf}}, c_k)$	$\cos(g_{\text{reg}}, c_k)$	$\cos(g_{\text{reg},k}, c_k)$
SDG 1	0.367	0.717	0.717
SDG 2	0.383	0.598	0.708
SDG 3	0.388	0.486	0.698
SDG 4	0.347	0.449	0.539
SDG 5	0.349	0.499	0.701
SDG 6	0.404	0.478	0.745
SDG 7	0.379	0.478	0.741
SDG 8	0.416	0.637	0.643
SDG 9	0.413	0.530	0.731
SDG 10	0.360	0.649	0.739
SDG 11	0.427	0.543	0.655
SDG 12	0.387	0.468	0.733
SDG 13	0.385	0.495	0.752
SDG 14	0.387	0.377	0.812
SDG 15	0.393	0.451	0.729
SDG 16	0.252	0.349	0.572
SDG 17	0.415	0.527	0.768

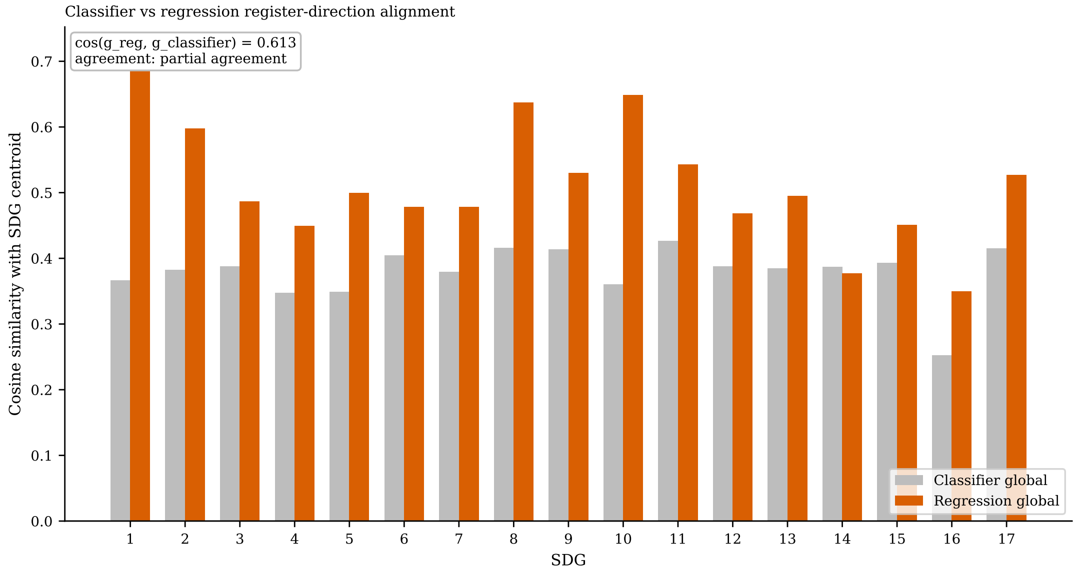


Figure 11. Classifier-derived versus regression-derived register alignment with the OSDG centroids. The annotation reports the cosine similarity between the global regression vector and the SDG-balanced classifier vector.

Afterword

This project emerged from a series of convergences rather than a single moment of design. I first encountered the Sustainable Development Goals early in my undergraduate studies. I later learned natural language processing during my master's. Over time, I noticed that the gap between research and policy is real, persistent, and often unspoken. Eventually, these threads converged.

To me, the connection felt natural, almost obvious. But that, I have come to realise, is precisely the point of education: not simply the accumulation of knowledge, but the gradual layering of perspectives until, at some point, they overlap. And when they do, something new becomes visible, something that, in hindsight, feels like it had always been there.

I came into this MSc believing, perhaps naively, that AI is going to be the game-changer for this world — or at least for a framework that tries to direct and measure how this world should be changed (the SDGs). In fact, that very belief was why I wanted to do this degree at all. And during my masters year, I learned a lot about how AI is actually used and talked about in research — practically, to change the world, not just as some grandiose ideology, but also in the very basic, mundane, daily tasks: irrigation, energy management. And I wondered: do policymakers and politicians talk about this too? Because I honestly believed, for some reason, that they do, and they should.

Turns out that last bit was not as formal as I thought.

The corpus asymmetry discussed in Section 5.5 captures this empirically. I had built the research corpus expecting it to mirror a policy world that also talks about AI *for* the SDGs. Instead, the policy documents turned out to be about AI or about the SDGs — the AI $\cap$ SDG intersection that felt natural to me was not yet naturalised in institutional discourse. The mismatch I found was not just between two corpora; it was between my mental model of the world and the world itself.

And maybe, just maybe, the ones to bridge that research-to-policy gap, specifically in AI-for-sustainability, are we who graduate from the MSc AI and Sustainable Development cohorts from UoB.

This, more than any method or result, is the real learning. I live my life, I observe things, I chose to do this MSc because I had an intuition, a belief, maybe even a dogma. I did the dissertation. I realized I might need to update my beliefs. On top of all the ML methods, the development theories, the AI applications — the real learning is the way I think.

I honestly could not stand the dryness of research. I want to express. I love to express, to make art, to tell stories, to share laughter. I believe that that is a strength I have to will navigate our changing world in the decades to come, to be the human who will be able to use AI wisely and, haha, sustainably.